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SPECIALIZATION PROJECT

Tracking Pedestrians in Oslo with Distributed Acoustic Sensing

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Abstract

In this project we use distributed acoustic sensing (DAS) to track marching bands in the 17th of May parade in Oslo. We employ an adapted version of the time series alignment technique called dynamic time warping (DTW), which we use to align DAS signals at subsequent locations. This allows us to estimate the position of pedestrian groups at different times along the monitored area. Thereafter, we define a statistical model combining the results from DTW and Gaussian random fields to model a velocity field of the parade, and improve the tracking obtained from DTW.

Using DTW and the combined model, we successfully track marching bands, producing results that are more accurate than simply guessing a constant walking speed of 1 m/s. These results show the potential of DAS to monitor pedestrian movement in urban areas. We also highlight shortcomings of the defined models, which in particular involves sensitivity to the choice of similarity measure. We found cross-correlation to be the most promising measure to locate pedestrians, but more work can be done in this area to provide more accurate tracking.

Preface

Working on this project was time-consuming, but very rewarding when things worked out. Some hours were spent doubting whether we would get the results we anticipated, but our efforts usually gave results in the end, much thanks to the expertise of my supervisor, Jo Eidsvik.

The results in this project were produced using Python, with DAS data stored in CSV files. Despite not being publicly available, implementation of the written code is carefully explained in text for the sake of reproducibility. As a part of the work process, some AI-tools were used, including GitHub Copilot for help with fast coding and ChatGPT for final spelling correction. Illustrations were made in Inkscape.

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1 Introduction

Distributed acoustic sensing (DAS) has a wide variety of applications, such as reservoir surveillance (Johannessen et al., 2012, Mateeva et al., 2014), monitoring railways, tunnels and pipelines (Zhang et al., 2021, Zhu et al., 2022), and whale tracking (Rørstadbotnen et al., 2023b). The technique utilizes Rayleigh backscattering of laser light in fiber optic cables to detect the position and intensity of acoustic events. Current technology for DAS can monitor a cable reliably over distances of up to 100 kilometers, and is remarkably robust, making the technique suitable for monitoring almost any environment (He and Liu, 2021).

An area where DAS yet has a large potential for development is in monitoring pedestrian movement in urban areas (Rørstadbotnen et al., 2023a). Pedestrian flow plays a critical role in city planning, public safety and transportation management, and is therefore an important field of research (United Nations Human Settlements Programme, 2013). Traditionally, this flow is monitored using methods like video surveillance and manual counting (Martani et al., 2017), which may face limitations regarding coverage, privacy concerns and scalability. DAS is a non-intrusive and scalable method which utilizes existing fiber optic cables to provide real-time monitoring, and as such it may prove useful for tracking pedestrian movements.

In this project, we explore the application of DAS technology to monitor pedestrian flow in Oslo during the annual celebration of Norway’s Constitution day, May 17th. During this event, a parade consisting of several schools and marching bands walk through the city, isolated from disturbances such as cars, trams and trains. A fiber optic cable which was previously used for commercial purposes was connected to a DAS instrumentation by Rørstadbotnen et al. (2023a), and used to obtain DAS data of the celebration in 2023. The cable coincides with key segments of the parade path, allowing us to test the use of DAS to monitor pedestrian flow along such segments. We are aiming to follow marching bands in the parade possibly moving with changing pace, by the use of the obtained DAS data.

In this project, we make two main contributions to DAS technology for pedestrian monitoring:

- **Dynamic Time Warping (DTW):** We develop a DTW-based method that uses cross-correlation between power spectral densities as a similarity measure, allowing us to track groups of pedestrians moving along a monitored area. DTW has frequently been used in areas such as speech recognition (Berndt and Clifford, 1994), data mining (Ratanamahatana and Keogh, 2004), and handwriting recognition (Efrat et al., 2007), but the adaption to DAS data for pedestrian monitoring is new.
- **Joint Velocity Warping:** We define a method that models pedestrian walking speeds as a Gaussian random field, enabling more consistent and realistic tracking of groups by conditioning on observed velocities from the entire parade. Warping methods in general are used in areas such as image processing (Glasbey and Mardia, 1998), where the goal is to match two images by warping, or “stretching”, certain parts of one image to better fit the other. We expand

these ideas to pedestrian movement, and iteratively warp a velocity field based on observed velocities, which we in turn use to track pedestrian groups based on their (simulated) walking speeds at different points in space and time.

Both contributions are based on large theoretical foundations in statistics and signal processing, and require deep understanding of the DAS data to develop. As such, a significant part of the work in this project is devoted to time-frequency analysis of the DAS data, where we find, explain and apply theory to the dataset.

In Section 2 we explain the principles of DAS in detail, and provide an overview of the scope of the project. Section 3 consists of data analysis, where we first focus on the time domain before we switch to spectral analysis to discover periodic patterns in the data. In Section 4 we introduce the concepts of DTW, and define an algorithm inspired by DTW to track marching bands of the Oslo parade through time and space. Finally, in Section 5, we define the joint velocity warping model, which we use to improve the results obtained from DTW.

2 Distributed Acoustic Sensing in Oslo

2.1 Principles of DAS

The DAS instrumentation (interrogator) located at one end of a DAS cable sends out pulses of laser light through a fiber optic cable. Due to a phenomenon known as Rayleigh scattering, parts of the light are reflected back to the instrumentation after colliding with impurities in the fiber (Zhu et al., 2020). When there is no strain in the cable, the backscattered light will return to the interrogator at specific time lags depending on the distance from the interrogator to the impurities. Vibrations caused by acoustic events such as trains, cars or people walking strain the cable at the locations near this activity, causing phase changes in backscattered light. This means that two subsequent laser pulses reflected in the same part of the cable return to the interrogator at different delays. Such a phase change is proportional to the length of the cable strain, and therefore the interrogator locates acoustic events by measuring phase changes in the backscattered laser light. The principles of DAS and Rayleigh scattering are illustrated in Figure 1.

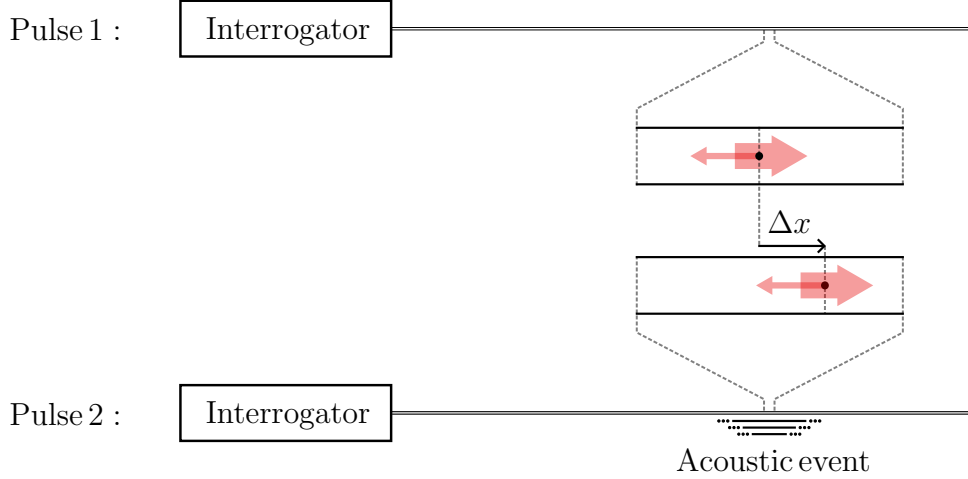


Figure 1: The interrogator sends out one laser pulse at a time. Rayleigh backscattered light from two different pulses return to the interrogator at different lags due to an acoustic event causing a strain in the cable, thus moving an impurity by a distance Δx .

2.2 Case Description and Data Acquisition

Testing the capabilities of DAS in monitoring pedestrian movement in urban areas can be challenging. People move in unpredictable ways, and larger acoustic events caused by cars, trains, and so on deafens the acoustic signals of pedestrians, making it hard to gain insights into how a fiber optic cable can be used for monitoring purposes (Brownjohn et al., 2004). For this project, however, we use data obtained by Rørstadbotnen et al. (2023a) from a fiber optic cable traversing Oslo centre, which was made available over the course of 11 days from May 11th to May 22nd 2023. On May 17th Norway celebrates its constitutional day, and therefore cars, buses,

trams and trains are not allowed to enter certain parts of Oslo city centre. Moreover, in celebration, a parade walks through the city. This opens up an opportunity to test the use of fiber optic cables to monitor pedestrians as we already know when, where and in what direction a large group of people will walk. This is essential information if we want to make any useful conclusions from the observations we make with the acquired data. Our case thus revolves around the use of DAS to monitor the movements of the parade in Oslo that occurred on May 17th 2023.

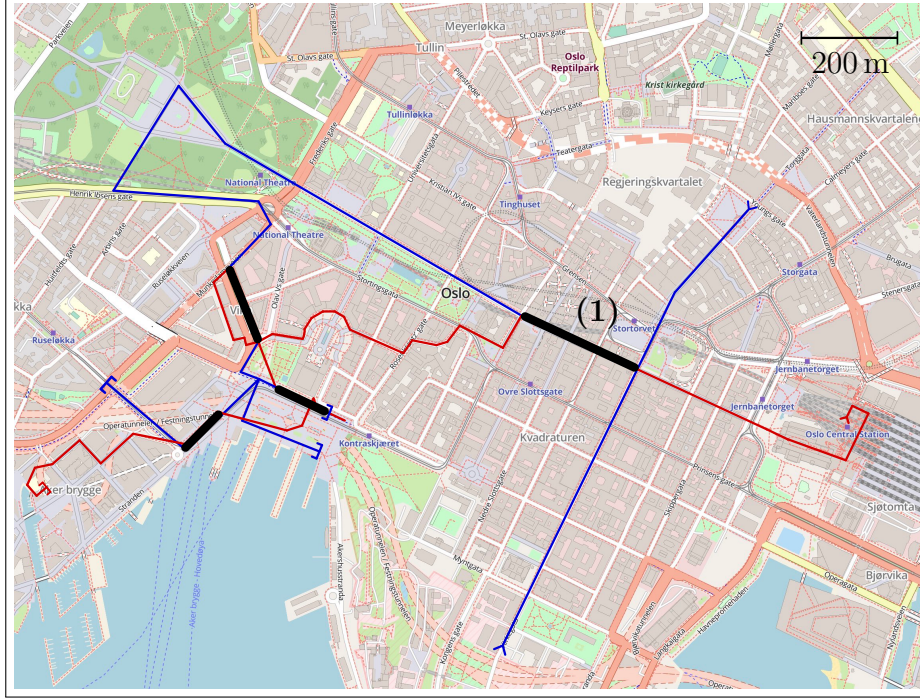


Figure 2: Map over Oslo with the parade path (blue line) and the cable trajectory (red line). Arrows are starting points, brackets are end points. In this project we limit our scope to Karl Johans gate, i.e. the black line (1), where the cable and parade path overlap. (MapTiler, 2024).

The parade path and the positioning of the fiber optic cable are shown in Figure 2. For this project we limit our scope to the data captured by receivers along the street highlighted by (1), which is part of the street called Karl Johans gate. It is the longest section where the parade path and the cable trajectory overlap, and all parts of the parade will walk past this area regardless of starting point. This makes the data from Karl Johans gate the most reasonable choice for statistical analysis of the DAS signal generated by the parade, and as such we will only focus on the data from this section.

In total the fiber optic cable traversing Oslo is 3.88 kilometres long. At the start of the cable there is an interrogator sending out laser pulses at a frequency of 2500 Hz. The interrogator used in this experiment register acoustic events at discrete locations, or “receivers”, each separated by 1.02 meters. Thus there are a little over 3800 receivers in total along the cable, of which we will use data from the 58 first receivers along Karl Johans gate. For instance, the first channel at Karl Johans gate will be given position $s = 0$, and is referred to as “Location 0”. In total the distance from first to last receiver is 58.14 meters.

3 Data Analysis of the DAS Dataset

Data analysis of the DAS dataset is divided into two sections. First we focus on the time domain. This involves exploratory data analysis and an introduction to autocorrelation functions, which we use to discover periodic patterns in the data. We also introduce the notion of cross-correlation as a similarity measure. Thereafter we analyze the data in the frequency domain to gain further insights into periodicity. Here we first look at the discrete Fourier transform, which we use to generate power spectral densities and spectrograms.

3.1 Temporal Analysis of a DAS Signal

3.1.1 Exploratory Data Analysis

We start by plotting the signals registered by the interrogator. Figure 3 shows approximately 5 hours of recorded strain from the first receiver on Karl Johans gate. Each sampling time is separated by 0.016 seconds, and the total signal length is $N = 1.134.692$ samples. Time is relative to 07:23:55 (UTC) on the morning of May 17th 2023. Note in particular the large spikes in strain highlighted by the first two stapled lines, which correspond to the times the King’s Guard and a police parade enters the monitored area. Also note the significant increase in nanostrain at the third stapled line near 1 hour and 15 minutes, which corresponds to the starting time of the parade. The final stapled line marks the (approximate) end of the parade.

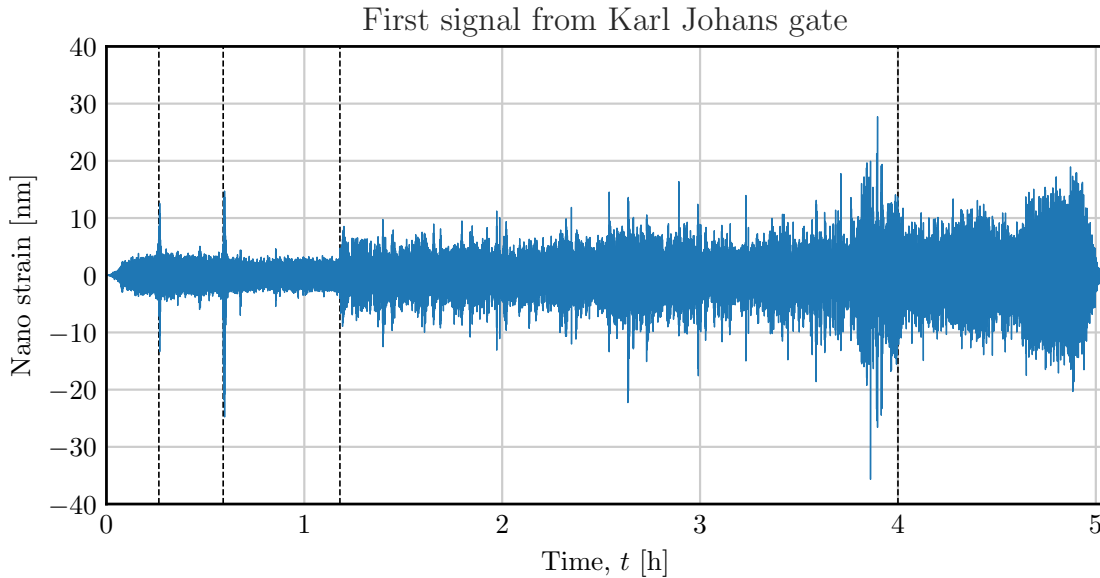


Figure 3: Recorded signal at the first location on Karl Johans gate, Location 0. The four stapled lines highlight (1) the King’s Guard entering the area, (2) police inspecting the parade path, (3) start of the parade, and (4) end of the parade.

Figure 4 displays the registered strain at all $M = 58$ locations. Also here, the

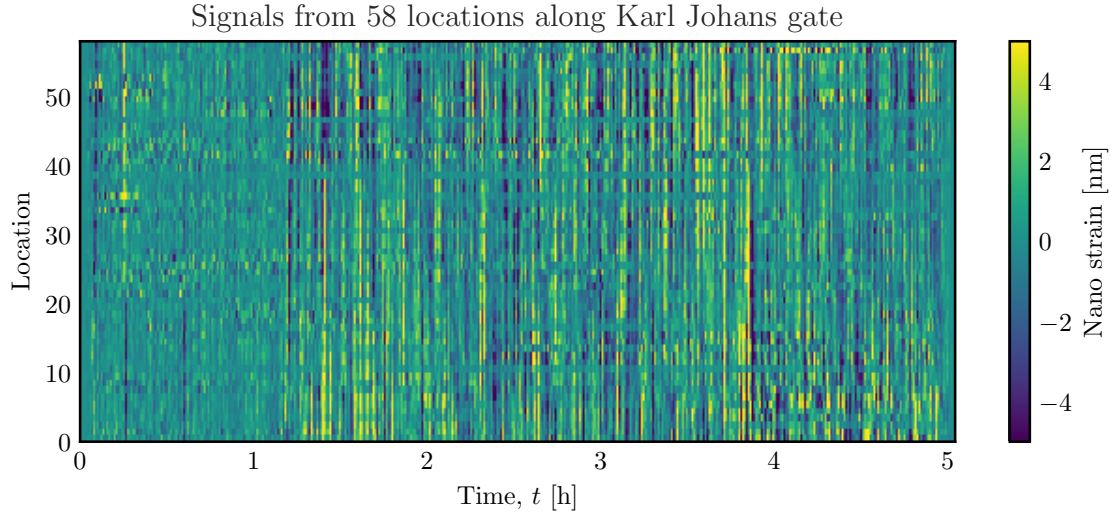


Figure 4: Strain from the 58 channels on Karl Johans gate.

events highlighted in Figure 3 are visible. Note for example the sudden increase in signal strength after 1 hour and 15 minutes, corresponding to the start of the parade. Moreover, we observe that the signal strengths vary slightly from location to location. For instance, one receiver near location 10 has an overall weak signal compared to the signals near location 50. This is likely caused by physical properties such as depth below surface, surrounding soil and impurities in the cable, which may vary from one location to another, thus affecting the signal strength.

We further compare the signals by assessing sample means and variances over the 58 locations, which we include in Figure 5. In the left display we observe that the means are zero, which is as it should be considering that the equilibrium of the cable is at zero strain. The right display shows that the variances are different, confirming the suspicion of location-dependent signal strength. The signal at location 10 has a low variance while the signal at location 49 has a very high variance, which is coherent with the observations we made from Figure 4.

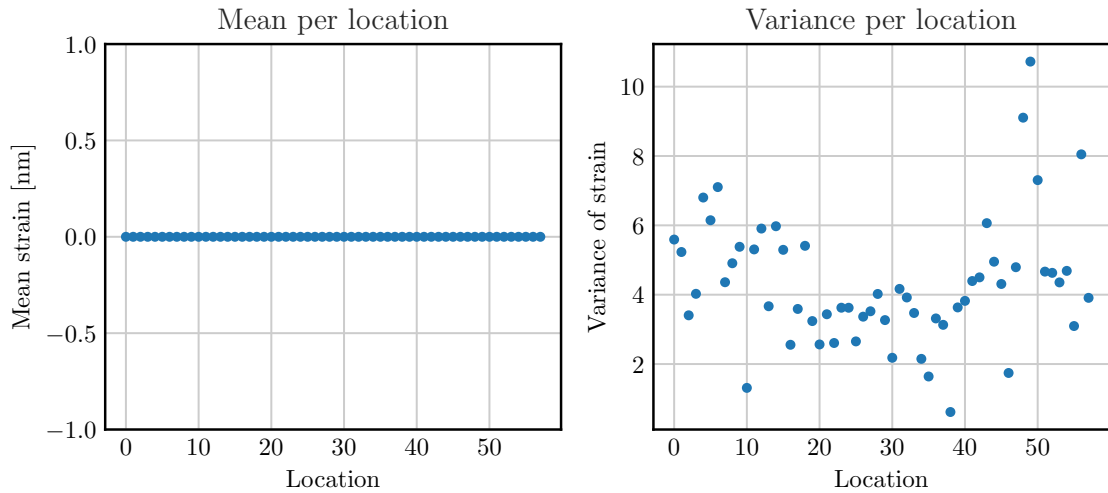


Figure 5: Mean and variance of each of the 58 signals.

3.1.2 Autocorrelation Functions

In this subsection we focus on temporal autocorrelation of a signal at a fixed location. Let $s \in \{0, \dots, 57\}$ denote location and t_n , $n = 0, \dots, N - 1$ denote the evenly spaced sampling times of the signals, with $t_0 = 0$. A signal $\{x(t_n; s)\}_{n=0}^{N-1}$ has a corresponding autocovariance function $C(\tau; s)$, which is obtained using

$$C(\tau; s) = \sum_{n=0}^{N-1} x(t_n; s)x(t_n + \tau; s), \quad \tau \in \{0, \dots, t_{N-1}\}. \quad (1)$$

Here, τ represents time lag, and can take the discrete time values from $t_0 = 0$ up to t_{N-1} . The signal is assumed to be zero if $t_n + \tau$ is greater than t_{N-1} . The autocovariance function is essentially a measure of how similar the signal is to itself when shifted by some lag, and as such it can be used to detect periodic patterns in the data. We use the function as a similarity measure at different lags, and to make it more interpretable for this purpose we normalize the function by dividing by the variance of the signal. From Figure 5 we know that the mean is approximately zero, and thereby the variance corresponds to the first value of the autocovariance function. This gives the autocorrelation function of the signal,

$$c(\tau; s) = \frac{C(\tau; s)}{C(0; s)}, \quad (2)$$

which takes values on the interval $[-1, 1]$.

We are not interested in the function at all time lags, as only the first few seconds should be sufficient to detect periodic patterns caused by pedestrian movement. We find the autocorrelation function for the first 4 seconds, and display it in Figure 6.

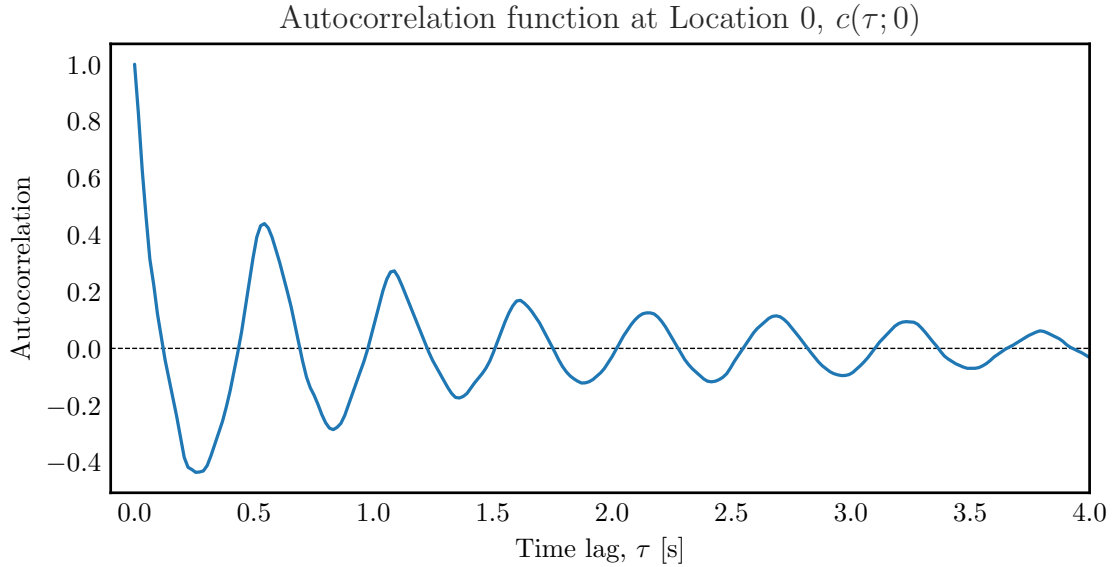


Figure 6: First 4 seconds of the autocorrelation function of the signal at Location 0.

The autocorrelation function in Figure 6 has the appearance of a damped periodic curve approaching 0. The period of the function is just above half a second, which

corresponds to a frequency of just below 2 Hz. The signal is mainly reflecting the parade, and their frequency matches well with the standard pacing frequency range of 1.2 to 2.4 Hz found by Živanović et al. (2010).

To obtain a more precise measure of the period we can fit a curve to the autocorrelation function and study its properties. Once again we are only interested in the first few seconds, which correspond to N_r time steps. Based on the appearance of the function in Figure 6, we fit a damped cosine curve of the form

$$f(\tau; \alpha, \beta, \gamma) = e^{-\alpha t^{1/\beta}} \cos(2\pi\gamma\tau). \quad (3)$$

The unknown parameters $\mathbf{\Gamma} = (\alpha, \beta, \gamma)$ must be determined using some feasible method, and since the function is non-linear in the parameters a reasonable approach is to choose the $\mathbf{\Gamma}$ that minimize the square residuals,

$$\mathbf{\Gamma}^* = \arg \min_{\mathbf{\Gamma}} \sum_{n=0}^{N_r-1} (c(\tau_n; s) - f(\tau_n; \mathbf{\Gamma}))^2. \quad (4)$$

The value of N_r depends on the number of observations within the time interval selected for fitting the function. In this case we fit the function using data from the first 4 seconds of time lag, thus we determine N_r accordingly.

Here, (4) is a non-linear least squares problem, which we solve numerically using the Levenberg-Marquardt optimization method as defined by Sauer (2014, p. 236). In particular, given an initial guess for the unknown parameters $\mathbf{\Gamma}_0$ and some fixed constant $\lambda \in [0.1, 100]$, we find the next value $\mathbf{\Gamma}_{k+1}$ by first solving for $\mathbf{v}_k$ in the equation

$$(\mathbf{A}^T \mathbf{A} + \lambda \text{diag}(\mathbf{A}^T \mathbf{A})) \mathbf{v}_k = -\mathbf{A}^T \mathbf{r}_k,$$

where $\mathbf{r}_k = \{(c(\tau_n; s) - f(\tau_n; \mathbf{\Gamma}_k))^2\}_{n=0}^{N_r-1}$ are the residuals, $\mathbf{A}$ is the Jacobian of the residuals at iteration k and λ is a regularization parameter. Then, our next guess for the unknown parameters is $\mathbf{\Gamma}_{k+1} = \mathbf{\Gamma}_k + \mathbf{v}_k$.

We apply the Levenberg-Marquardt method with initial guess $\mathbf{\Gamma}_0 = (1, 1, 1)$ and $\lambda = 100$ to fit the damped cosine curve (3) to the autocorrelation function from Location 0. The optimal parameters $\mathbf{\Gamma}^*$ are

$$\mathbf{\Gamma}^* = (1.45, 2.25, 1.84). \quad (5)$$

In Figure 7, we plot the fitted curve together with the autocorrelation function to evaluate the fit visually. The curve $f(\tau; \mathbf{\Gamma}^*)$ appears to fit the autocorrelation function well, at least when it comes to the periodic component. The amplitudes at the second and third peaks are slightly off, and as time increases the amplitude of the fitted function seem to diminish slower than that of the autocorrelation function. Still, the period of the fitted function seems to match that of the data autocorrelation quite well. This frequency is a measure of the dominating frequency in the data, and thus it can be used as an estimate for the typical walking frequency of the parade at a specific location. Thus, this analysis indicates that $\omega = 1.84$ is the dominating step frequency at Location 0.

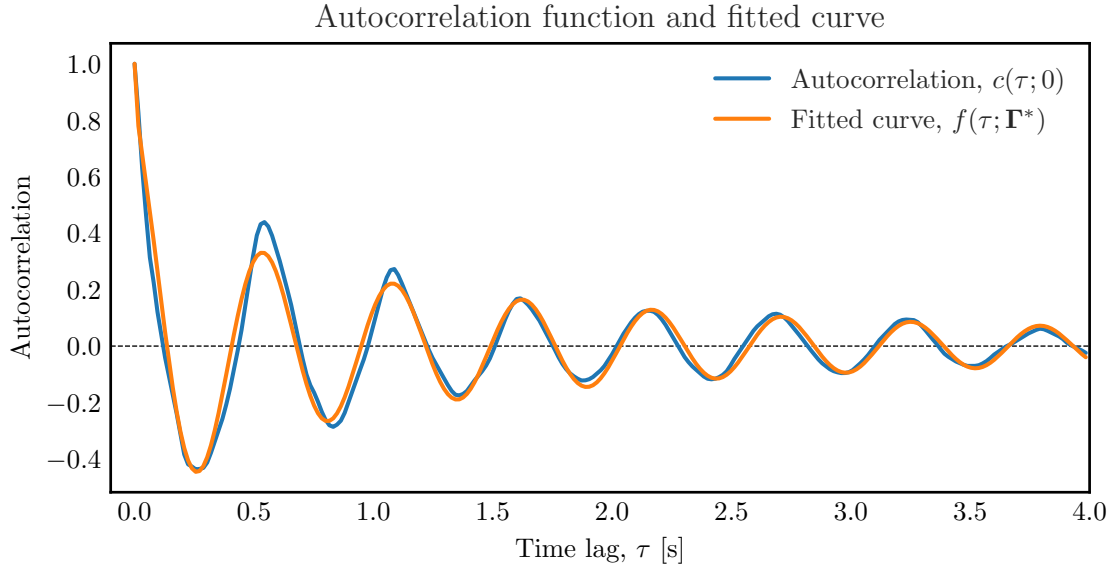


Figure 7: Autocorrelation of the signal from Location 0 on Karl Johans gate, along with a curve fitted using the Levenberg-Marquardt method.

3.1.3 Cross-correlation

Cross-correlation is essentially a generalization of the autocorrelation function, in the sense that an autocorrelation function is the cross-correlation between a signal and itself. Given two signals $\{x(t_n)\}_{n=0}^{N-1}$ and $\{y(t_n)\}_{n=0}^{N-1}$, their cross-covariance is

$$R(\tau) = \sum_{n=0}^{N-1} x(t_n)y(t_n + \tau), \quad \tau \in \{0, \dots, t_{N-1}\}. \quad (6)$$

We normalize (6) to obtain the cross-correlation,

$$r(\tau) = \frac{R(\tau)}{\max_{\tau} |R(\tau)|}, \quad (7)$$

which takes values on the interval $[-1, 1]$. By evaluating the cross-correlation at 0 time lag we obtain a similarity measure between the two time series, where a value close to one indicates high similarity. This similarity measure plays an important role in the models we define in Sections 4 and 5, and examples of its use are included there.

3.2 Spectral Analysis of a DAS Signal

Pedestrians usually walk with a somewhat constant frequency. For this reason, spectral analysis of the DAS data may prove useful. The idea behind spectral analysis relies heavily on the Fourier transform, which converts the signal from the time to the frequency domain, allowing us to study periodic patterns in the data. In this section we first introduce a formal definition of the Fourier transform for discrete signals. We next define two useful tools for data analysis: power spectral densities and spectrograms.

3.2.1 The Discrete Fourier Transform

The signals used for analysis in this project are time series, each sampled at evenly spaced locations along the DAS cable. At a fixed location s we let $\{x(t_n; s)\}_{n=0}^{N-1}$ denote the time series obtained by the receiver at that location. The discrete Fourier transform (DFT) of such a signal is defined as

$$\hat{x}(\omega_k; s) = \sum_{n=0}^{N-1} x(t_n; s) e^{-2\pi i \omega_k t_n}, \quad k = 0, \dots, N-1, \quad (8)$$

where N is the number of samples in the signal. This is an important tool for spectral analysis (Kreyszig, 2011, p. 530). It allows us to observe the signal in the frequency domain, which is particularly useful when we assume the data to have periodic patterns, as is the case for pedestrian movement.

The frequencies ω_k depend on the sampling rate of the data. Since we use a sample rate $\omega_s = 62.5$ Hz, we can only recognize periodic patterns up to a frequency $\omega_{\max} = \omega_s/2 = 31.25$ Hz, known as the Nyquist rate (Landau, 1967). The full frequency spectrum of the DFT therefore contains N evenly distributed frequencies on the interval $[0, 31.25]$ Hz. For our purpose of walking pattern characterization only positive frequencies within a certain range, say $\omega \in [0, 10]$ Hz, are of interest, so from now on we only consider the frequencies within this interval. The DFT of the signal from Location 0 on Karl Johans gate is displayed in Figure 8.

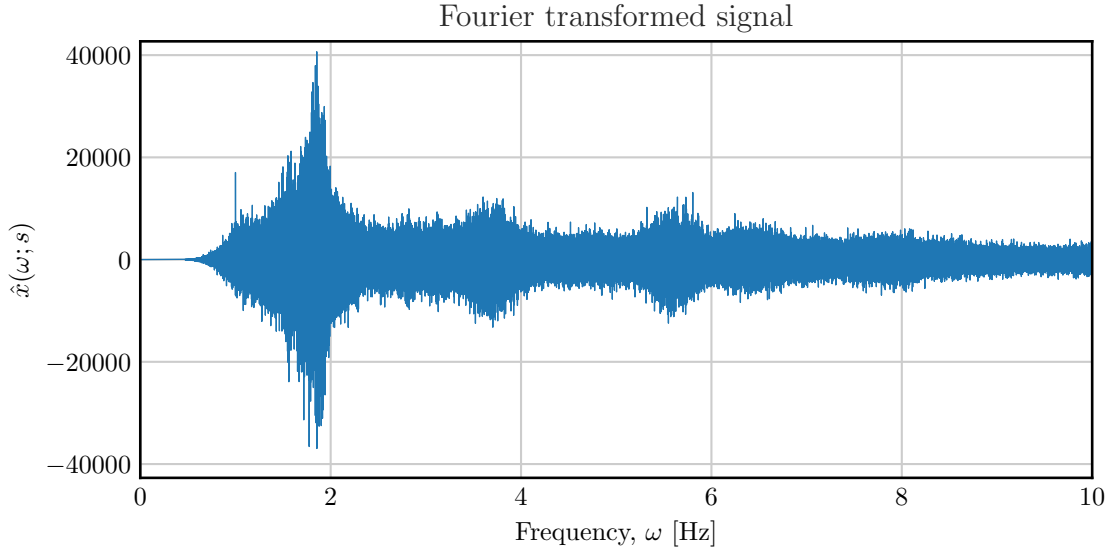


Figure 8: The DFT associated with the frequencies $\omega \in [0, 10]$ of the signal from Location 0.

The most dominant frequency of the DFT shown in Figure 8 appear to be around $\omega^* = 1.8$ Hz. This indicates that pedestrians move at almost two steps per second. Notice also how the Fourier transform has evenly occurring diminishing peaks at approximately $\omega = 2\omega^*$, $\omega = 3\omega^*$, and so on. This is a known property of the Fourier transform, and happens because the original signal is decomposed into a sum of sinusoidal waves with amplitudes relative to the presence of this frequency

in the original signal. Since sinusoidal waves with frequencies $\omega = 2\omega^*$, $\omega = 3\omega^*$ and so on also will occur with frequency $\omega = \omega^*$, we see peaks at these values in the Fourier transform (Kreyszig, 2011). This phenomenon has been observed in several other papers on pedestrian movements as well, such as ones by Živanović et al. (2010) and Brownjohn et al. (2004). We call the most dominant frequency ω^* for the fundamental mode, and the diminishing peaks for higher-order modes.

3.2.2 Power Spectral Densities and Spectrograms

Once we have calculated the discrete Fourier transform of the signal we can find the power spectral density (PSD),

$$S(\omega_k; s) = |\hat{x}(\omega_k; s)|^2, \quad k = 0, \dots, N-1, \quad (9)$$

which shows the power of the signal as a function of frequency. The PSD is similar to the Fourier transform in shape, but the squared absolute value gives us a measure of total volume of the signal, regardless of the amplitude being positive or negative. In the case of the DAS cable, negative and positive amplitudes both mean the same thing, i.e. a straining of the cable as a result of some acoustic event. This makes the PSD a more useful tool for analyzing pedestrian movement.

The PSD of a discrete signal is easily influenced by random noise, resulting in occasional spikes that do not make sense when compared to neighbouring values. To even out such spikes we use Hamming smoothing, where we replace each value $S(\omega_k; s)$ with a weighted average of the neighbouring values, over a specified window centered at ω_k and with size N_H (Harris, 1976). In particular, each value $S(\omega_{k+\ell-\lfloor \frac{N_H}{2} \rfloor}; s)$ within the window is multiplied by

$$h(\ell) = 0.54 - 0.46 \cos \frac{2\pi\ell}{N_H - 1}, \quad \ell = 0, \dots, N_H - 1, \quad (10)$$

and as such the smoothed PSD is defined by

$$\tilde{S}(\omega_k; s) = \sum_{\ell=0}^{N_H-1} S(\omega_{k+\ell-\lfloor \frac{N_H}{2} \rfloor}; s) h(\ell), \quad k = 0, \dots, N-1. \quad (11)$$

In edge cases where the frequencies in the smoothing window fall outside the range $[0, 10]$ Hz, we assign a value of 0 to those frequencies instead of including values outside the specified window. This will not affect our analysis as the values of the PSD will be close to 0 here, anyways.

We need N_H to be small compared to the total number of frequencies N , so the appropriate choice of N_H will depend on N . In Figure 9 we plot the PSD and the smoothed PSD together, where we set $N_H = 2.000$ to clearly demonstrate the effects of Hamming smoothing. This number is small compared to $N \approx 1.130.000$.

Figure 9 displays the PSD of the signal along with its smoothed version. The smoothed curve do not share the jagged movement or occasional spikes of the non-smoothed PSD. Notice for example how there is a large spike in the PSD around 1

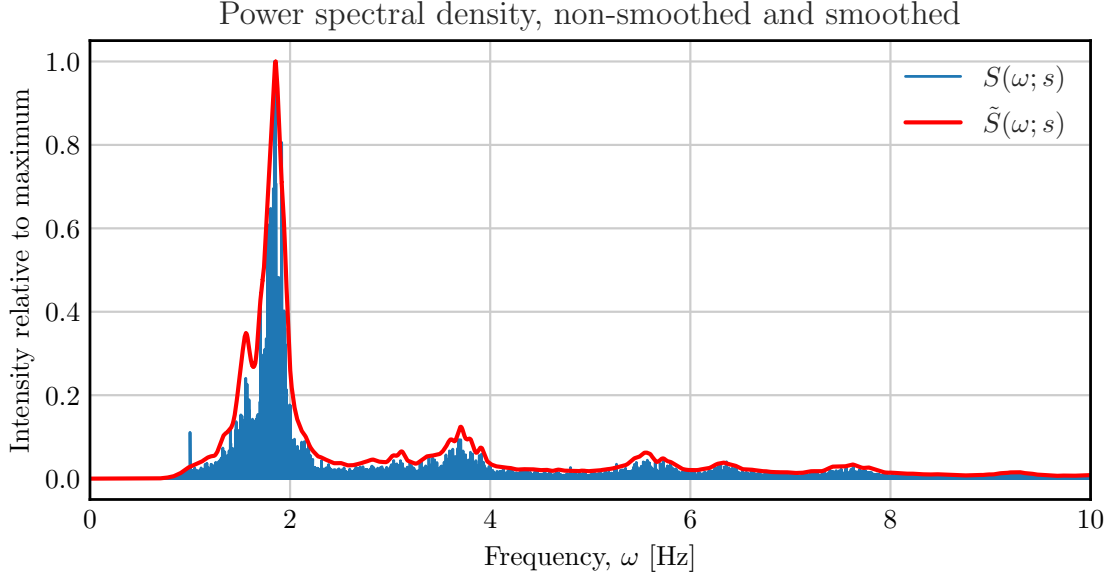


Figure 9: The PSD $S(\omega; s)$ (blue) and the smoothed PSD $\tilde{S}(\omega; s)$ (red) of the signal from Location 0. Both curves are divided by their respective maximum value to show similarity in shape.

Hz which is evened out in the smoothed PSD. This means the smoothing is working as intended. Also note the presence of the higher order peaks that we observed in Figure 8.

The PSD in Figure 9 has been calculated using an entire 4-hour long signal. Using a method known as short time Fourier transform (STFT), however, we can find an approximation to how the PSD changes over time. The idea is to divide the entire signal into relatively small windows of size N_H , each centered at a unique time step and overlapping by a certain amount, and calculate the PSD using the data in each window. The window size is for this purpose kept at $N_H = 2048$ data points, and since each sample is taken 0.016 seconds apart, this equals about 32.8 seconds. The overlap between each window is 90% of the window size, and works as a way of smoothing the PSDs over time. Consequently, we are left with 5573 time steps after performing STFT, each separated by 3.28 seconds. Moreover, to ensure the PSD centered at a certain time point is not too heavily affected by the values closer to the edge of the window, we once again make use of the Hamming window defined in (10). Each data value within the window is weighted according to the Hamming window, so that the values closer to the middle have greater importance when calculating the PSD. After calculating the PSDs of all such small time windows, we plot them as a function of time, in what is called a spectrogram. The PSD centered at a certain time point t is then plotted at time t .

Figure 10 shows the spectrogram corresponding to the signal from Location 0. Here, we notice some evident signs of pedestrian movement. During the first hour the main parade has not yet started, and thus there is little activity to be seen. There are occasional events, however, one of these being a police parade a little more than 30 minutes into the data sampling. This appears as a clear increase in amplitude over a

short period of time. The fact that the short time Fourier transform captures single events such as the police parade is an indication that $N_H = 2048$ is not too large.

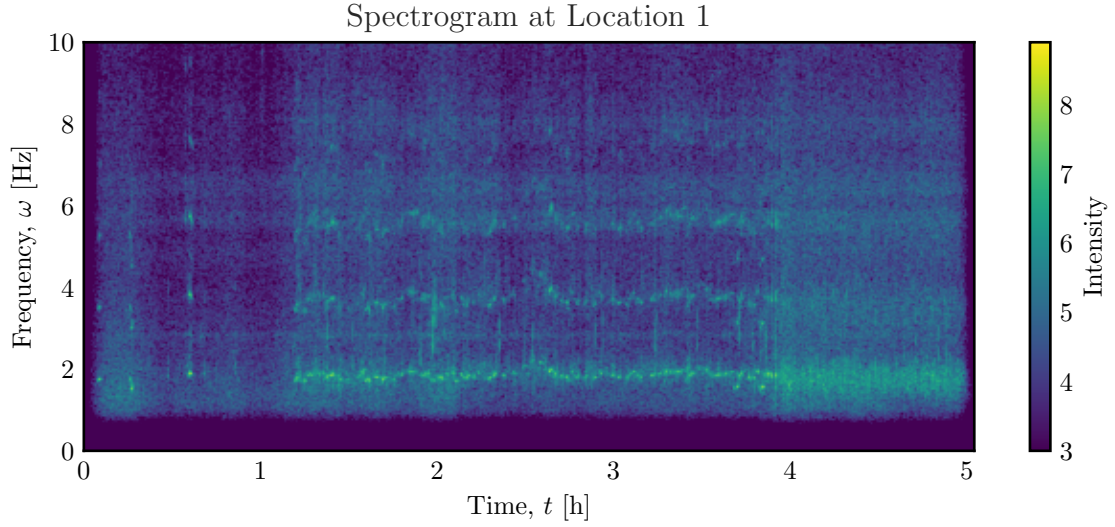


Figure 10: Spectrogram of the signal from Location 0.

Notice also how the spectrogram in Figure 10 clearly captures the fundamental mode around 2 Hz as well as the higher-order modes that we saw in Figure 9, despite each window only containing about 30 seconds of data. This indicates that the choice of window size is sufficiently large. Moreover, the spectrogram is able to capture changes in the fundamental mode over time, which occur when a group of pedestrians with a unique frequency pass by the receiver. For instance, at approximately 2.5 hours, there is a slight increase in the fundamental mode, which may be marching bands walking at a higher frequency than other parts of the parade.

4 Dynamic Time Warping of DAS Signals

Dynamic Time Warping (DTW) is a method of aligning two time series where similar events occur at shifted time steps (Hale, 2013). This idea is illustrated in Figure 11. In the case of pedestrian movements registered by the DAS cable, similar events will be registered by different receivers as pedestrians that pass a receiver at location s at time t will pass another receiver at location $s + \Delta s$ at a later time, $t + \Delta t$. This is assuming that all pedestrians move in one direction along the cable, which is the case for the parade monitored on May 17th in Oslo. DTW allows us to find a one-to-one relation between values of two signals, which theoretically means that we can track pedestrians at one location and time to any subsequent location and time.

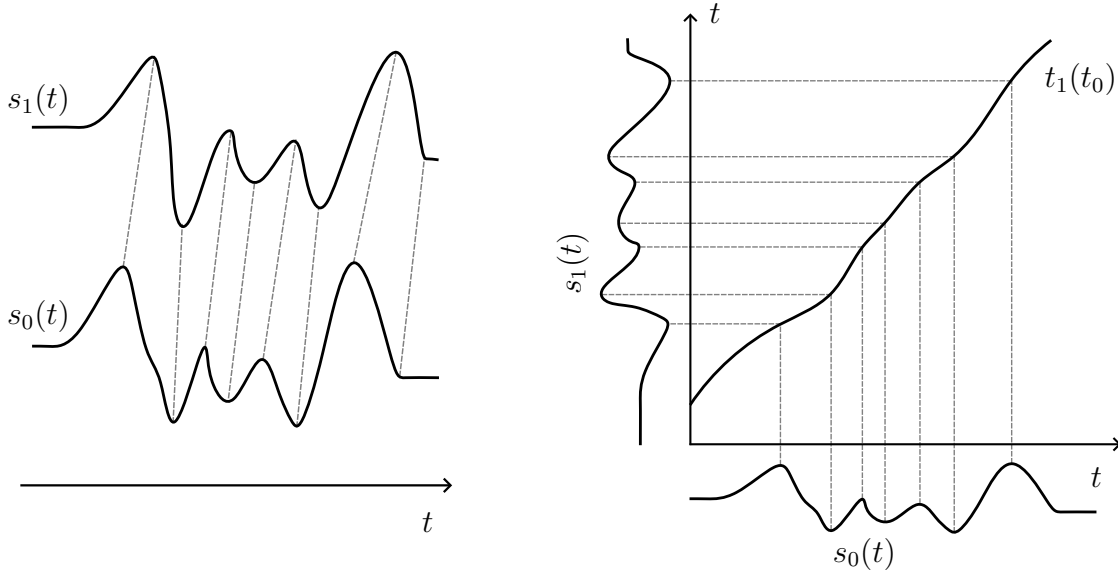


Figure 11: The idea behind DTW. Two signals are similar, but at different lags (left display). DTW can be used to find the optimal time-alignment of such signals (right display).

In this section we first define a new, altered version of the DTW algorithm, adapted to the use of tracking pedestrians with DAS data. Then we introduce cross-correlation as a similarity measure between PSDs, which we incorporate in the DTW algorithm to track pedestrians from the first location at Karl Johans gate to the last.

4.1 The DTW Algorithm

The algorithm we define in this subsection is a two-step method. Our goal is to find an optimal path through space and time for a group of pedestrians entering Karl Johans gate, in the sense that the path leads to minimal loss, which we will explain further in a later subsection. The first part of the algorithm is called the accumulation step, and the second is the backtracking step. The implementation of these is shown in Algorithm 1, and we explain each step thoroughly in separate subsections.

Algorithm 1 Dynamic Time Warping

▷ Accumulation step

```
for  $s = 0, \dots, M - 1$  do
     $\triangleright$  Iteratively build accumulated loss  $d(t, s)$ 
    for  $t = 0, \dots, N - 1$  do
        if  $s = 0$  then
             $d(t, 0) \leftarrow \frac{1}{3} \sum_{\delta=1}^3 e(t, 0, \delta)$   $\triangleright$  Accumulated loss at  $s = 0$ 
        else
            if  $t = 0$  then
                 $d(0, s) \leftarrow d(0, s - 1) + e(0, s, 3)$   $\triangleright$  Accumulated loss at  $t = 1$ 
            else
                
$$d(t, s) \leftarrow \min \begin{cases} e(t - 1, s, 1) + d(t - 1, s) \\ e(t - 1, s - 1, 2) + d(t - 1, s - 1) \\ e(t, s - 1, 3) + d(t, s - 1) \end{cases} \quad \triangleright t, s > 0$$

            end if
        end if
    end for
end for
```

▷ Backtracking step

```
 $\mathbf{u}_1 \leftarrow (\arg \min_{t \in T} d(t, M - 1), M - 1)$   $\triangleright$  Minimal loss at last position
 $\mathbf{U}[1] \leftarrow \mathbf{u}_1$ 
 $i \leftarrow 1$ 
while  $s_i > 0$  do
     $\triangleright$  Find path along lowest accumulated loss
     $\mathbf{u}_{i+1} = (t_{i+1}, s_{i+1}) \leftarrow \arg \min_{(t,s) \in \mathcal{P}_i} d(t, s), \quad \mathcal{P}_i = \{(t_i - 1, s_i), (t_i - 1, s_i - 1), (t_i, s_i - 1)\}$ 
     $\mathbf{U}[i] \leftarrow \mathbf{u}_i$ 
     $i \leftarrow i + 1$ 
end while
```

4.1.1 Accumulation Step

In the accumulation step we calculate the (minimum) accumulated loss along all paths in time from the first to the last location, given the constraint that the parade do not move backwards in time or space. To do this we need a measure of loss at each time and location, or what we call alignment errors $e(t, s, \delta)$, indicating the error we make by deciding on a certain path from one point in space and time to the next. The alignment error is a function of time t , location s and direction δ . The last parameter δ is included because of our choice of alignment error, which we will come back to in a later subsection. The idea is that the error we make on deciding on a certain path depends on both what point we move to, and what point we come from, hence the need for a directional parameter. Due to the constraints that the parade does not move backwards in location or time, the only directions the parade can move in the (t, s) -plane is to the right, diagonally and up. We assign the values $\delta = 1$, $\delta = 2$ and $\delta = 3$ to these three directions, respectively, as illustrated in Figure 12.

Using the alignment error we calculate the accumulated loss $d(t, s)$ at each location and time. Here we let N denote the number of time steps and M the number of

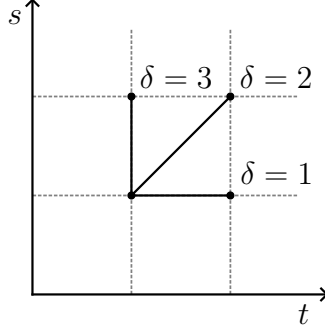


Figure 12: Directions associated with $\delta \in \{1, 2, 3\}$ in the accumulation step.

locations. Note that from here on we redefine N to be the number of time steps after we have performed STFT, i.e. $N = 5573$, as explained in Section 3.2.2. For each time step $t = 1, \dots, N - 1$,

$$d(t, 0) = \frac{1}{3} \sum_{\delta=0}^3 e(t, 0, \delta), \quad (12)$$

$$d(t, i) = \min \begin{cases} e(t-1, i, 1) + d(t-1, i) \\ e(t-1, i-1, 2) + d(t-1, i-1) \\ e(t-1, i-1, 3) + d(t-1, i-1) \end{cases} \quad \text{for } i = 1, \dots, M-1. \quad (13)$$

We do not have a measure of the error associated with arriving at $s = 0$. Instead we assume that the average of the alignment errors in all three directions can be used as an approximation, which gives (12). Then, at each time step for the next location, we find the accumulated loss as the minimal sum of relevant alignment errors and accumulated losses. Hence, we calculate the accumulated loss to reach a point from each of the three possible directions, separately, and choose the lowest value. For each direction, the accumulated loss is the sum of the accumulated loss at the previous point and the alignment error we make by moving from the previous point in the specific direction.

In (12) and (13) we have not considered the case when $t = 0$. Here we assume that the only possible path in the (t, s) -plane is straight up, which we do to make sure the algorithm runs smoothly. From the spectrogram in Figure 10 we already know that there is no parade movements recognizable near $t = 0$, so it does not make any difference in the optimal paths we find for the actual parade. The iterative calculation of $d(t, s)$ is illustrated in the left display of Figure 13.

Finally, we are left with a matrix of accumulated loss to reach each location at any time point, which we need for to perform the backtracking step.

4.1.2 Backtracking Step

The idea behind the backtracking step is to start with the minimal accumulated loss at the last location $s = M - 1$, and iterate backwards. Here we need to specify the time interval $T = [t_{\text{start}}, t_{\text{end}}]$ in which we wish to find an optimal path since we may want to find several paths using the same accumulated distance matrix, but at

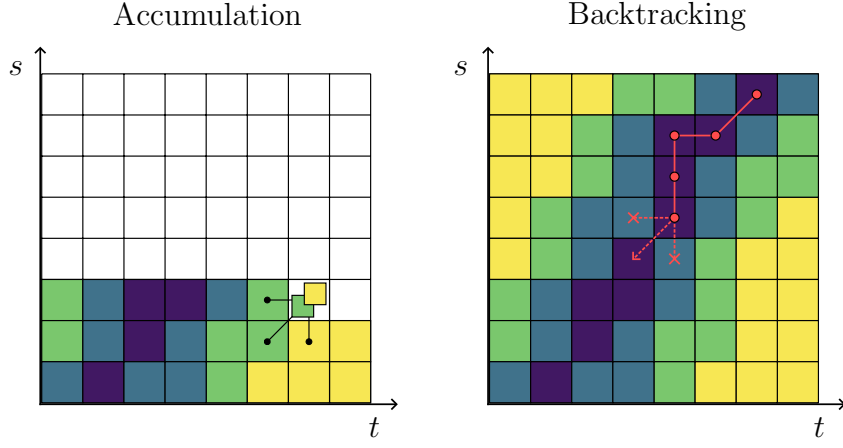


Figure 13: We build the accumulated loss matrix iteratively based on the minimal sum of local loss and alignment error (left display). Then we find the optimal time alignment of subsequent locations by matching minimal values, thereby also finding the optimal path (right display).

different time steps. Let $\mathbf{U} \in \mathbb{R}^{p \times 2}$ denote the optimal path in time and space from the last location to the first, that is,

$$\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_p), \quad \mathbf{u}_i \in \{(t, s) : t \in T \text{ and } s = 0, \dots, M-1\}. \quad (14)$$

The number of points p in the optimal path may vary depending on the time interval T . The starting point of the optimal path minimizes the accumulated loss at the final location,

$$\mathbf{u}_1 = \left(\arg \min_{t \in T} d(t, M-1), M-1 \right). \quad (15)$$

Continuing from this, the next point in the optimal path is the adjacent point where the accumulated loss is the lowest given that we do not move forward in time or location,

$$\mathbf{u}_{i+1} = \arg \min_{(t,s) \in \mathcal{P}_i} d(t, s), \quad \text{for } i = 1, 2, \dots, \quad (16)$$

where $\mathcal{P}_i$ denotes the three possible next points from $\mathbf{u}_i = (t_i, s_i)$,

$$\mathcal{P}_i = \{(t_i - 1, s_i), (t_i - 1, s_i - 1), (t_i, s_i - 1)\}. \quad (17)$$

The backtracking iteration is stopped once we reach $s = 0$, at which point we have found the optimal path from the last location to the first in the sense that accumulated loss is the lowest along this path. Thereby we have also found the most likely pedestrian path from $s = 0$ to $s = M-1$ given the assumed alignment errors. The backtracking step is illustrated in the right display of Figure 13.

An issue with the optimal path obtained from the two steps defined here is that we allow the path to be at different locations during the same time step, which obviously cannot be the case for the actual parade path that we are trying to model. This must be allowed since we use discrete time steps, and thereby we do not find the exact time at which the parade enter a certain location, but rather the time which

is closest. In case the time steps of the alignment errors are large compared to the distance between two receivers, we must allow the algorithm to assign several locations to the same time step. Therefore, to make the optimal path we obtain from the DTW algorithm more realistic, we remove the points where the optimal path takes the same time value as in a previous point. This ensures that the optimal path never attains “infinite speed”, thereby giving a more realistic estimation of the actual pedestrian path. Altering the optimal path like this is done after Algorithm 1 has been run.

4.2 Choice of Alignment Errors

To find the most probable path of the parade we need the alignment errors to be low at the points in space and time where pedestrians are likely to be, and high where they are less likely to be. Thus, to find a reasonable measure of alignment error, we first make assumptions on how we can detect whether or not pedestrians are likely to be at a specific point in space and time. In this subsection we first look at how PSDs can be used for this purpose, before we introduce similarity between PSDs and use this to define a reasonable alignment error.

4.2.1 Similarity Between PSDs as Basis for Alignment Errors

For the case of pedestrian movement we hypothesize that certain pedestrian groups, such as marching bands, can be uniquely identified by the signal obtained from them walking over a receiver. Assuming this to be true, if a group of pedestrians is at a certain point in space and time, the PSD at this point should be similar to the PSD at the next point the parade moves to, and differ slightly from PSDs at points outside the parade path. When we talk about the PSD at a certain time point we refer to the PSD obtained using short time Fourier transforms with a Hamming window of size $N_H = 2048$, as discussed in Section 3.2.2. This means that if we know the current location of the parade, we can find the next location by measuring the similarity of the PSD at this location and all possible next locations, and find the one with the best match. Moreover, if there is no parade present at a certain point, the recorded signal will only be noise, and as such the PSD should not be correlated with the PSD at a nearby point. For this reason similarity between PSDs is a reasonable basis for alignment errors. Next, we need a way to measure this similarity.

4.2.2 Cross-correlation to Measure Similarity between PSDs

An intuitive approach to compare PSDs is to use simple difference methods, such as difference between maximum amplitude or even the area between densities. After attempting these, however, we found that the amplitudes of the PSDs may vary by quite a lot from one location to another, rendering such similarity measures useless. Thus we need a measure that does not rely solely on the amplitudes of the PSDs, but also on the shape. This makes cross-correlation a reasonable choice. As such,

to compare two PSDs, we calculate the cross-correlation between the two as defined in (7), and evaluate at $\tau = 0$ to obtain a measure of similarity.

In our case, the cross-correlation must be defined as a function of several factors. The DTW algorithm requires a measure of similarity that depends on time t , location s and direction δ , so to use cross-correlation as a basis for alignment error it should be a function of t , s and δ . Thus, $r(t, s, \delta)$ will from here on refer to the cross-correlation between the PSD at the point (t, s) and the PSD at the neighboring point associated with the direction δ .

If we have N time steps and M locations the calculated cross-correlation matrix $\mathbf{R}$ is of size $M \times N \times 3$. To plot correlation matrices, we take the sum of the cross-correlation in the three directions, and use the normalized sum to represent the cross-correlation at each point (t, s) . We do this for the $M = 58$ locations on Karl Johans gate and $N = 5573$ time steps, and display the resulting cross-correlation in Figure 14.

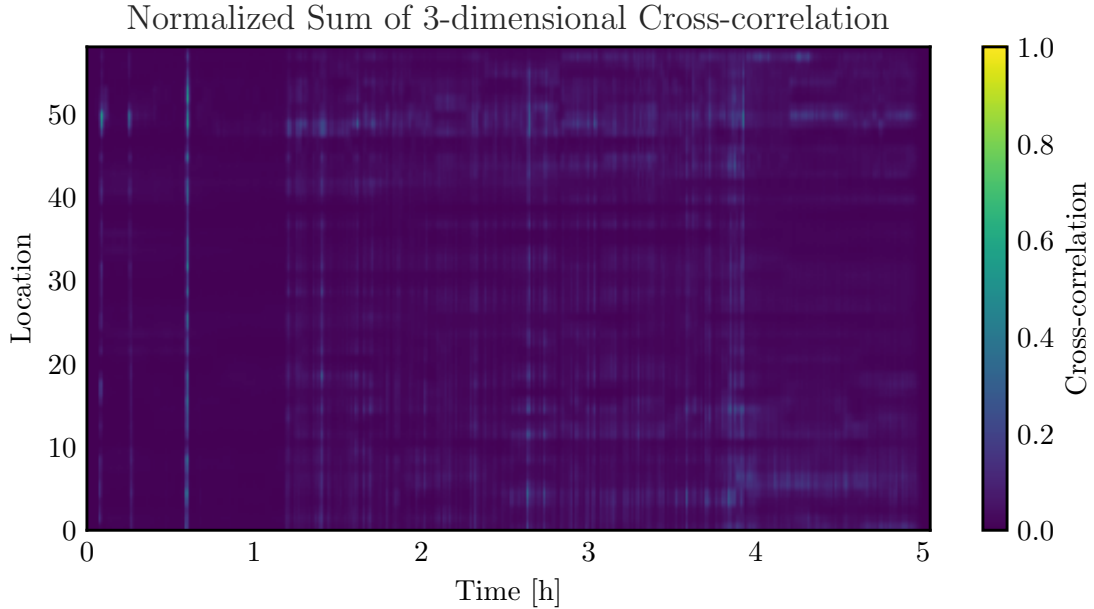


Figure 14: Normalized sum of cross-correlation over the 3 directions $\delta = 1$, $\delta = 2$ and $\delta = 3$.

From Figure 14 it appears like cross-correlation works well as a similarity measure. It seems to be able to capture different parts of the parade as it moves forward, represented by clear lines of high correlation. In particular, there appear to be 3 clear events in the first hour, followed by a large increase in the number of events just after 1 hour. This is a good match with the known starting time of the parade.

If we zoom into some visible events, we notice that they are diagonal, indicating that the cross-correlation picks up on the pedestrians moving forward in time and space (Figure 15). The slopes of these lines are measures of walking speed, as it shows the estimated location of a part of the parade per time step. In the left display, for instance, the pedestrians appear to spend just below 1 minute to walk about 60 meters, i.e. a walking speed of around 1 m/s, which seems reasonable. In the right

display, the walking speed appear to be around 2 meters per second.

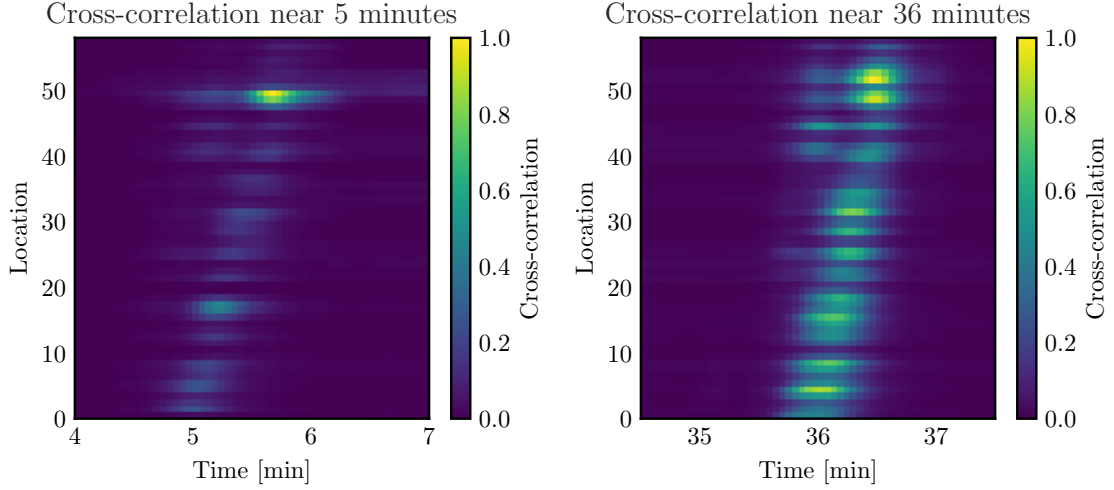


Figure 15: Zoomed-in view of the cross-correlation matrix near 5 and 36 minutes.

Another interesting observation from Figure 15 is that the cross-correlation is location-dependent. The correlation at location 10, for instance, is very low compared to the correlation near location 50, for both plots. This is due to the varying signal strength at each location, which we discussed in Section 3.1.1. To limit such factors from influencing the optimal path found using the DTW model, we smooth the cross-correlation matrix using a Gaussian kernel with higher variance along the diagonal axis corresponding to a speed of 1 m/s. The choice of 1 m/s as the axis is based on an article by Rørstadbotnen et al. (2023a), where they found 1 m/s to be the typical walking speed of the parade along Karl Johans gate. In addition to limiting physical factors from affecting the optimal path, Gaussian smoothing with a diagonal axis may make it easier to distinguish between different parts of the parade walking in close succession. A group of pedestrians may for instance enter Karl Johans gate before the previous section has had time to leave, resulting in diagonal lines in the cross-correlation that are grouped close together, possibly interfering with the DTW algorithm. We display the smoothed versions of the correlation lines from Figure 15 in Figure 16, and evaluate the smoothing visually.

Figure 16 shows the two smoothed correlation lines normalized using their respective maximum values. We observe that the variation with respect to location is lower, which indicates that the smoothing works as intended. Near the edges the smoothed cross-correlations take slightly lower values, which is due to the implementation of the smoothing. We convolve the original matrix with the Gaussian kernel, and along the edges we pad the correlation matrix with zeros, thus resulting in lower values at these points.

4.2.3 Defining the Alignment Error based on Cross-Correlation

The alignment error is a measure of the error we make when deciding on a path from one point to a neighboring point. The cross-correlation we calculated in the

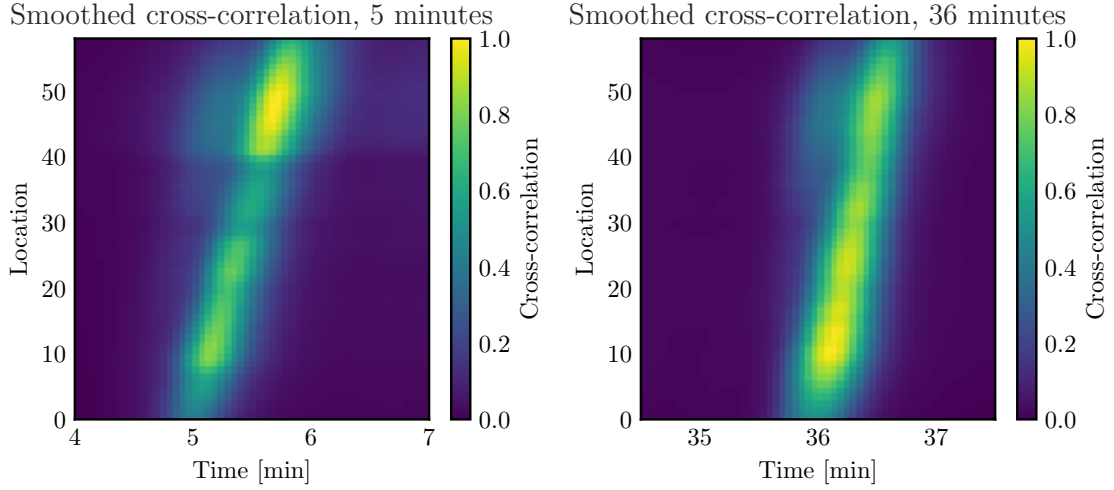


Figure 16: Zoomed-in view of the smoothed cross-correlation matrix near 5 and 36 minutes.

previous subsection is essentially the opposite, i.e. a measure of local similarity, in terms of similar PSDs at neighboring points. Therefore we define the alignment error as

$$e(t, s, \delta) = 1 - r(t, s, \delta), \quad (18)$$

such that the lowest alignment error is associated with the highest local cross-correlation, and vice versa.

4.3 Applying the DTW Algorithm to DAS Data

Now we apply the DTW algorithm based on cross-correlation to the DAS data from Oslo. First we perform the accumulation step, i.e. calculate the accumulated loss matrix. Then we apply the method to the two isolated events shown in Figure 16, before we extend the method to high-amplitude time steps of the parade.

4.3.1 Calculating the Accumulated Loss

Using the alignment error we calculate the accumulated loss $d(t, s)$. We do this as explained in the accumulation step of Algorithm 1, i.e. by adding up the (minimal) alignment error along each possible path. The resulting accumulated loss matrix is shown in the left display of Figure 17. In the right display we show a row-normalized version. Since the accumulated loss has gradually increasing values at each location, the low-loss areas are not as visible as in the row-normalized version, where there are several clear lines. These clear lines remind of those in the cross-correlation matrix. Note for instance the start of the parade near 1 hour and 15 minutes.

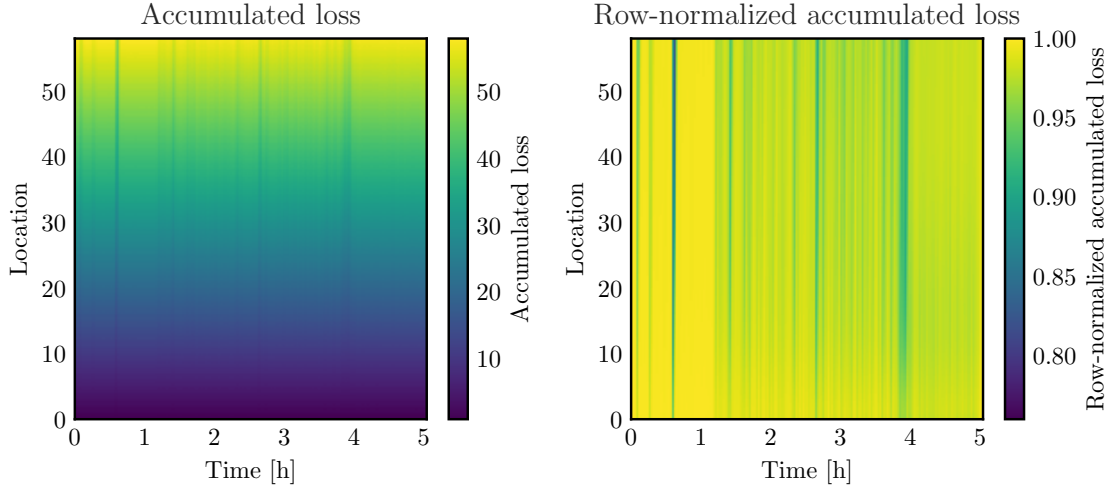


Figure 17: Accumulated loss and row-standardized accumulated loss.

4.3.2 DTW of Isolated Events

To find the optimal path for a specified event we fix the interval $T = [t_{\text{start}}, t_{\text{end}}]$ such that it captures the entire event. For instance, for the event in the left display of Figure 16 we could choose $T = [4, 7]$ minutes. In this subsection we do this manually, i.e. based on visual inspection of Figure 16, and in the next subsection we formalize the method of choosing an interval. The optimal paths are shown in Figure 18, along with the row-normalized distance matrix zoomed in at the corresponding time intervals. We normalize each row to make the differences in accumulated loss more visible, as the accumulated loss will increase for each location, making subtle differences hard to spot.

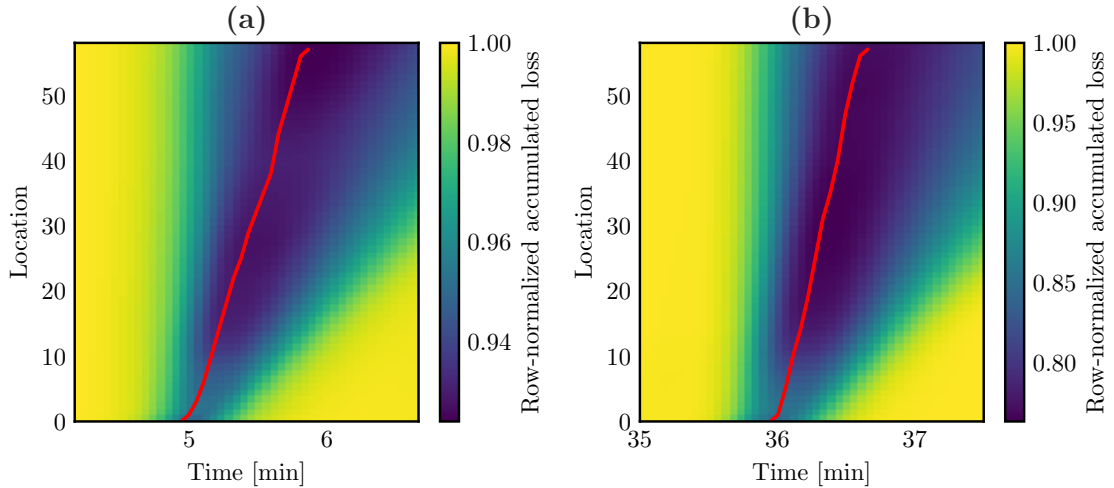


Figure 18: Row-normalized accumulated loss with optimal paths (red lines) found using the DTW algorithm. **(a)** Event at 5 minutes, and **(b)** event at 36 minutes.

In Figure 18 we display examples of optimal paths found using the adapted DTW model based on cross-correlation. The paths appear to follow the minimum accumulated loss at all locations, which is what we intended. Moreover, it appears like

the average slopes are about 1 m/s and 2 m/s for the paths in the left and right display, respectively, which is coherent with the observations we made from Figures 15 and 16. The accumulated loss also make sense visually, as it is low where we would expect the parade to be, and high where we would not. A potential issue we can observe regarding the accumulated loss is that the relative differences between the lowest and the highest values, are quite small. For example, in the left display of Figure 18 the minimal accumulated loss is around 93% of the maximum, which can be seen from the values of the color bar. The two events we have used for analysis were isolated and quite “obvious” by visual inspection of the correlation matrices, and as such the DTW algorithm may struggle with less obvious events. We will analyze the use of the DTW algorithm for such events in the next subsection.

4.3.3 DTW of the Parade

When we performed DTW in the previous subsection we looked at isolated events that occurred before the start of the actual parade. Now we use the same method on the actual parade, i.e. from approximately 1 to 4 hours into the signal. To do this, we first need a method of finding the times when a pedestrian group enters Karl Johan. After finding all these time points, we can find the optimal path through space and time for each interesting time point, as we did in a couple examples in the previous subsection. This would give a warping of signals at all time steps where the signal is a result of pedestrian movement.

To find time steps where groups of pedestrians enter Karl Johans gate we use the average cross-correlation at the first location. This is equivalent to $1 - d(t, 0)$. The idea here is to find all maxima of $1 - d(t, 0)$ that are at least Δt seconds apart and are above a threshold $d_{\min}$ relative to the highest peak. The separation time Δt is set to 60 seconds, and is meant to ensure that two close minima that are actually caused by the same acoustic event are considered the same event. The threshold $d_{\min}$ is determined so that we obtain a reasonable number of peaks. From the official website of Oslo municipality in 2023, we find that 110 schools and 88 marching bands partook in the parade. Here we make the assumption that marching bands give the most prominent peaks since members of a marching band walk with the same frequency, thus resulting in a more powerful signal obtained by the interrogator. Consequently we set $d_{\min}$ such that we obtain approximately 88 peaks in the parade, and attempt to track all marching bands. Figure 19 displays the (normalized) average cross-correlation at $s = 0$, where the 87 highest maxima are highlighted. This result was obtained using $\Delta t = 60$ seconds and a relative threshold $d_{\min} = 0.09$.

Many of the peaks in Figure 19 make sense, but there are also a quite a few that are not so obvious. Notice for instance the peak highlighted by a green cross near 1.5 hours, which barely surpass the threshold. We expect the DTW algorithm to struggle at such points, and perform well at the higher peaks.

Now we run the DTW algorithm for all obtained time points, and look at some examples. To do this we iterate over all peaks, and for each set t_{start} to a value slightly lower than that of the corresponding peak, and t_{end} sufficiently large to capture the entire event, say $t_{\text{end}} = t_{\text{start}} + 80$ seconds. Figure 20 displays the DTW

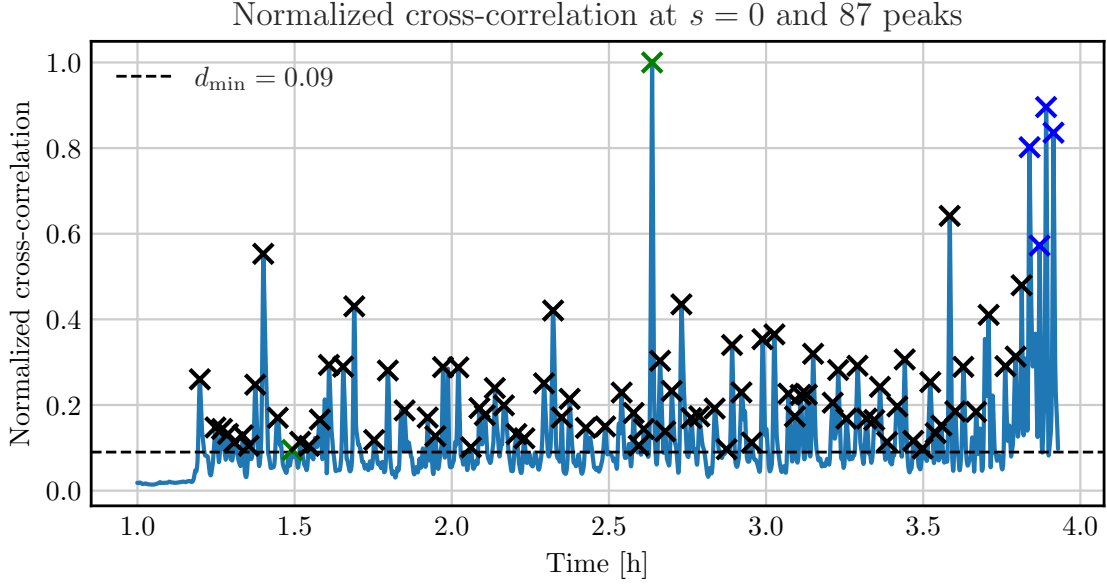


Figure 19: Normalized cross-correlation at $s = 0$ (blue line) and the 87 highest peaks detected with $\Delta t = 60$ seconds and $d_{\min} = 0.09$. Green and blue crosses represent the time steps used for analysis in Figures 20 and 21, respectively.

paths found at the highest and lowest peaks, which are colored green in Figure 19. Figure 21 shows paths that are close together in a time period of high activity, corresponding to the peaks colored blue in Figure 19.

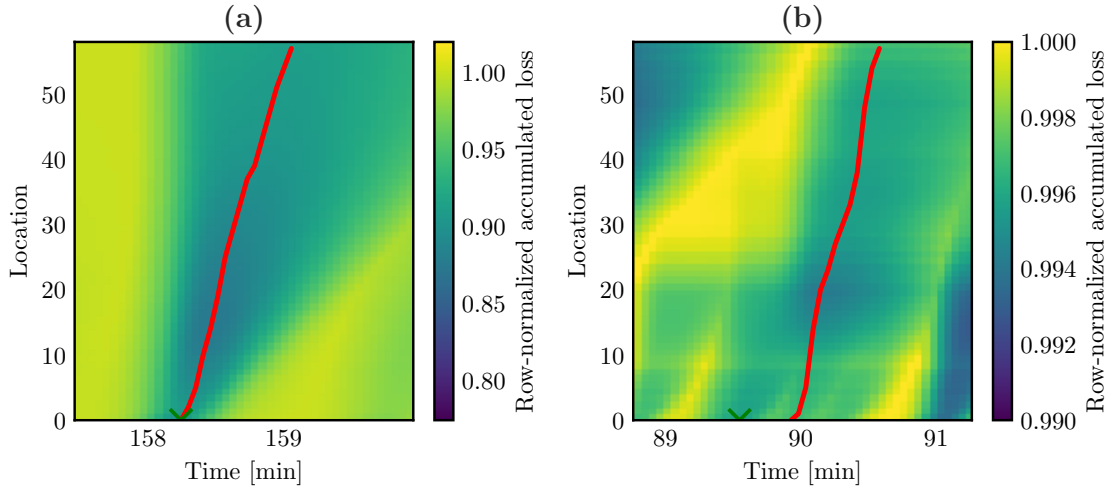


Figure 20: DTW paths found at **(a)** the highest and **(b)** the lowest peak, marked green. The peaks are also marked by green crosses in Figure 19.

The accumulated loss in the left display of Figure 20 is very similar to those of the isolated events in Figure 18, and the optimal path found is reasonable, indicating a walking speed of about 1 m/s. The path found in the backtracking step also ends up at the peak. In the right display, the path also seem somewhat reasonable, but the accumulated loss appears to be affected by nearby events. This makes sense as the row-normalized accumulated loss at all points of the path is above 0.99. As a

result, the path does not return to the peak. This supports our suspicion that the DTW algorithm performs better at finding the optimal path for events with high amplitudes, that stand out from nearby events.

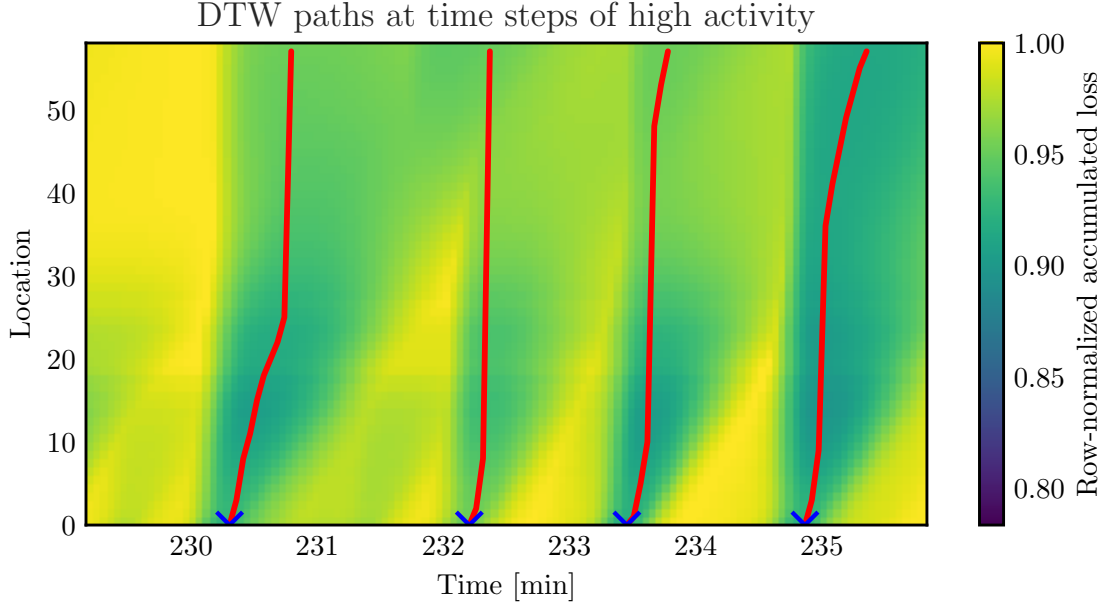


Figure 21: DTW paths found at four consecutive peaks at time steps of high activity. The peaks are marked by blue crosses in Figure 19.

Figure 21 shows four DTW paths found near the end of the parade, a time period where the overall activity is high. This can be seen from the high levels of cross-correlation around this time, displayed in Figure 19. Here the DTW algorithm does not find paths that seem likely, in the sense that the slopes of the four paths seem to be unreasonably high. For instance, the second path from the left is nearly vertical in the (t, s) -plane. This may be a result of a smaller event in between 230 and 232 minutes causing higher values of cross-correlation at the last locations near 232 minutes, thus affecting the choice of starting point in the backtracking step. As a result, the path quickly reaches the end of its time interval T . This is a limitation of the algorithm, showing that acoustic events easily are affected by other events. This, in turn, affects the accuracy of cross-correlation as a similarity measure, which causes the DTW algorithm to fail.

4.4 Applying Constraints on the DTW Lines

We want the optimal paths we obtain to be realistic, and as such we filter out the DTW lines that at any point suggest unreasonably high walking speeds, say $v > v_{\max} = 3$ m/s. We calculate the walking speed based on the slope of the DTW lines. After applying a maximum speed cap of 3 m/s, and removing all lines that do not satisfy this constraint, we end up with 34 lines satisfying the constraint, and 53 that do not. In Figure 22 we display the average speed per location with standard deviation obtained from these 34 DTW lines.

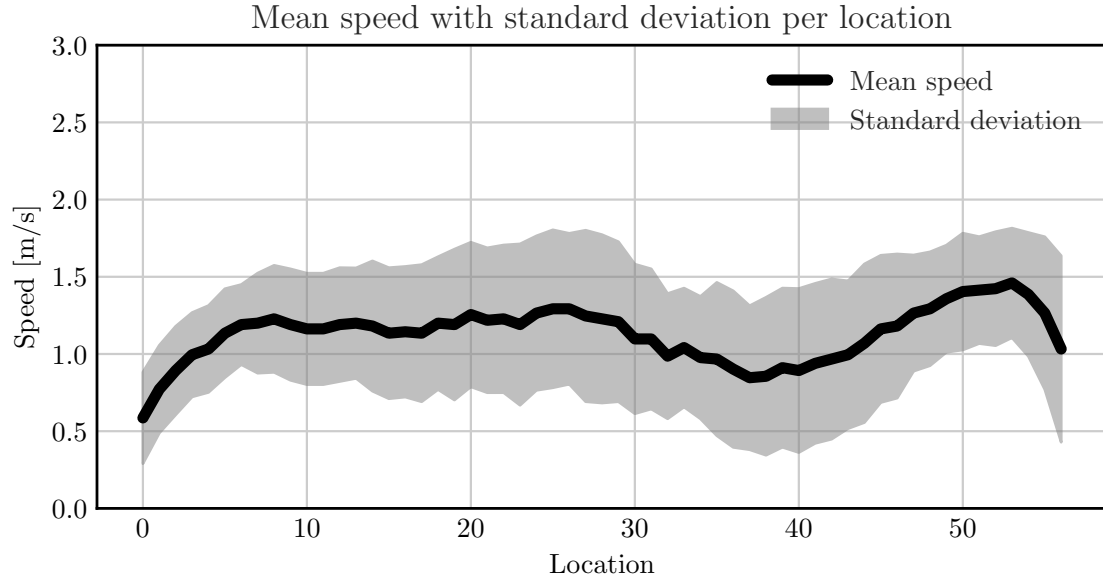


Figure 22: Average speed and sample deviation calculated from the 34 DTW lines satisfying $v < 3$ m/s.

The mean speed at each location shown in Figure 22 appear to fluctuate around 1 m/s, which is coherent with what was found by Rørstadbotnen et al. (2023a). The speed at the beginning of the monitored area is relatively low, at just above 0.5 m/s, and quickly increase to just above 1 m/s. Around location 40 the speed dips, and at the end of the monitored area the speed quickly declines.

5 Modeling a Velocity Field of the Parade

In the previous section we found that the adapted DTW model struggles to provide reasonable paths for groups that are close together, or that cause weak signals. In this section we attempt addressing these shortcomings by the means of a joint velocity warping model. The idea is to model the velocity of the parade as a Gaussian random field, condition this field on a set of estimated paths (hereby called “observations”), and use this conditional field to generate new, possibly improved observations, where information is shared between all paths. We then use cross-correlation to measure whether or not a path is improved, and repeat the process with updated observations. We aim to improve the 53 unreasonable DTW paths by using information contained in the 34 reasonable ones, and thereby obtain a better tracking of all marching bands in the parade.

In this section we first summarize the notation we use and the assumptions we make to define the model. Thereafter, we introduce the algorithm we have developed, and finally present our results.

5.1 Notation and Model Assumptions

Let $\mathbf{V}$ denote a velocity field, which is an $M \times N$ matrix containing velocities at each of the $M = 58$ locations and $N = 5573$ time steps. We define $\mathbf{v}_{\text{obs}} \in \mathbb{R}^{MP}$ to be the vector containing MP observed velocities, one per monitored location for each of the $P = 87$ observed paths. Each observation is made at a unique point in time and space, which we store in the matrix $\mathbf{A}_{\text{obs}} \in \mathbb{R}^{MP \times MN}$. That is, $\mathbf{A}_{\text{obs}}$ is a sparse matrix containing 1s at the coordinates of each observation, and 0s elsewhere. For example, if we have observed a velocity v at path p and point (j, k) , then $(\mathbf{A}_{\text{obs}})_{pM+j, jN+k} = 1$ and $(\mathbf{v}_{\text{obs}})_{pj} = v$.

The statistical aspects of the velocity warping model are based on the Bayesian framework. We assume that $\mathbf{V}$ is the matrix form of a sample $\mathbf{v} \in \mathbb{R}^{MN}$ from the prior distribution of the velocity of the parade, and the observed velocities $\mathbf{v}^{\text{obs}}$ is a sample from the likelihood. Based on these two, we can sample a conditional velocity vector $\tilde{\mathbf{v}}$, which is a sample from the posterior distribution, i.e. a sample which incorporates the information in the observed velocities, $\mathbf{v}_{\text{obs}}$. We make the following assumptions on the prior, likelihood and posterior distributions.

- The prior distribution of $\mathbf{v}$ is $N(\boldsymbol{\mu}, \boldsymbol{\Sigma}_T)$. Consequently, $\mathbf{V}$ is a Gaussian random field, which entails that any collection of points from $\mathbf{V}$ is a multivariate normal random variable (Abrahamsen, 1997). We let $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_T$ denote the mean and covariance matrix of $\mathbf{v}$, where we assume $\boldsymbol{\mu} = \mathbf{1}$ to be the mean velocity. This is based on the observed velocities in Section 4.4. $\boldsymbol{\Sigma}_T$ is assumed to depend on the distances Δs_j and Δt_k between two points in the plane that are separated by j and k indices in space and time, respectively. Two sets of two points that are separated by the same distance in both directions are assumed to have the same covariance. Thereby $\boldsymbol{\Sigma}_T$ is circulant, and contains at most MN unique entries, one for each of the MN possible relative positions of two

points in $\mathbf{V}$. It is, however, of size $MN \times MN$, which is not computationally feasible to work with as $MN \approx 320.000$ in our case. By utilizing the circulant properties, we can instead use a smaller matrix $\Sigma \in \mathbb{R}^{M \times N}$ containing only the MN unique entries. We define

$$\Sigma = \left\{ \sigma^2 \exp \left\{ -\frac{1}{2} \frac{(\Delta s_j)^2 + (\Delta t_k)^2}{\phi^2} \right\} \right\}_{j,k=0}^{M-1, N-1}, \quad (19)$$

where σ^2 is the variance and ϕ is a parameter determining the degree of spatial and temporal correlation. Based on the observations in Section 4.4 we set $\sigma^2 = 0.25$. It makes sense for the velocity to be notably correlated at least 10 meters and 10 seconds apart, thereby we set $\phi = 10$.

- The observed velocity vector $\mathbf{v}_{\text{obs}}$ is a sample from the likelihood, $N(\mathbf{A}_{\text{obs}}\mathbf{v}, \mathbf{P})$, where we assume a prior velocity field $\mathbf{v}$ has been observed. The covariance matrix $\mathbf{P} \in \mathbb{R}^{MP \times MP}$ contains values from Σ corresponding to the positions of the observations, which are stored in $\mathbf{A}_{\text{obs}}$.
- The posterior distribution is thereby a normal distribution, $N(\boldsymbol{\mu}_{\mathbf{v}|\mathbf{v}_{\text{obs}}}, \mathbf{P}_{\mathbf{v}|\mathbf{v}_{\text{obs}}})$ from which we can sample efficiently without having to estimate the mean $\boldsymbol{\mu}_{\mathbf{v}|\mathbf{v}_{\text{obs}}}$ or covariance matrix $\mathbf{P}_{\mathbf{v}|\mathbf{v}_{\text{obs}}}$ (Rue and Held, 2005). This sampling is explained in Section 5.2.2.

We further assume that cross-correlation is a reasonable measure of the loss $\mathcal{L}_{\text{obs}}$ associated with a set of observed paths. Consequently, the criterion for a path to be considered good is the same as it was in the DTW model, i.e. the higher cross-correlation, the better. What makes the obtained paths unique in this case, however, is that generating them from a velocity field naturally encourages velocities to be reasonable. Additionally, the randomness that comes with generated velocity fields allow the algorithm to discover paths that move across low-correlation regions, potentially finding high-correlation regions that are separated from the original area. The DTW-algorithm cannot find such paths as it always traverse locally based on minimal loss, and is therefore prone to inconsistencies in the cross-correlation. This idea is illustrated in Figure 23.

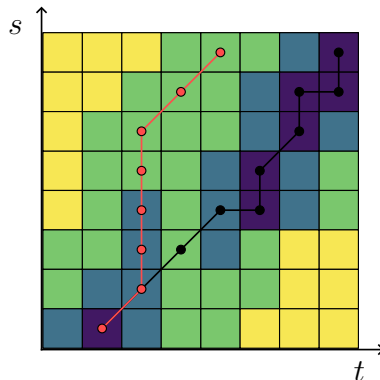


Figure 23: DTW finds an unreasonable path (red line) due to inconsistencies in the cross-correlation and thereby also in the accumulated loss. Simulating from velocity fields may yield a better path in such cases (black line).

5.2 The Joint Velocity Warping Algorithm

Now we introduce the steps in the joint velocity warping algorithm. This involves the simulation of a velocity field, conditioning on observations, and accepting new simulated paths. The general procedure is shown in Algorithm 2, and we explain the theory behind each step in separate subsections.

Algorithm 2 Joint Velocity Warping Algorithm

```

 $\mathbf{A}_{\text{obs}}$   $\leftarrow$  observed paths and velocities  $\triangleright$  Initial observations
 $\mathcal{L}_{\text{obs}}$   $\leftarrow$  loss( $\mathbf{A}_{\text{obs}}$ )  $\triangleright$  Loss along each initial path
Sample  $\mathbf{V}$   $\sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}_T)$   $\triangleright$  Velocity field, Section 5.2.1
 $\tilde{\mathbf{V}}$   $\leftarrow \mathbf{V} \mid \mathbf{A}_{\text{obs}}, \mathbf{v}_{\text{obs}}$   $\triangleright$  Conditional velocity field, Section 5.2.2

for  $b = 1, \dots, B$  do  $\triangleright$  Update lines based on loss, Section 5.2.3
   $\mathbf{A}_{\text{new}}, \mathbf{v}_{\text{new}}$   $\leftarrow$  SimulateLines( $\tilde{\mathbf{V}}$ )
   $\mathcal{L}_{\text{new}}$   $\leftarrow$  loss( $\mathbf{A}_{\text{new}}$ )

  for  $p = 1, \dots, P$  do  $\triangleright$  Iterate over all  $P$  paths
    if  $\mathcal{L}_{\text{new}}[p] < \mathcal{L}_{\text{obs}}[p]$  then  $\triangleright$  Update  $\mathbf{A}_{\text{obs}}, \mathbf{v}_{\text{obs}}$  and  $\mathcal{L}_{\text{obs}}$  if loss is reduced
       $\mathbf{A}_{\text{obs}}[p]$   $\leftarrow \mathbf{A}_{\text{new}}[p]$ 
       $\mathbf{v}_{\text{obs}}[p]$   $\leftarrow \mathbf{v}_{\text{new}}[p]$ 
       $\mathcal{L}_{\text{obs}}[p]$   $\leftarrow \mathcal{L}_{\text{new}}[p]$ 
    end if
  end for

  if change in  $\mathcal{L}_{\text{obs}} = 0$  then  $\triangleright$  Sample new velocity if loss does not change
    Sample  $\mathbf{V}$   $\sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}_T)$ 
  end if

   $\tilde{\mathbf{V}}$   $\leftarrow \mathbf{V} \mid \mathbf{A}_{\text{obs}}, \mathbf{v}_{\text{obs}}$   $\triangleright$  Re-condition velocity field on new  $\mathbf{A}_{\text{obs}}$  and  $\mathbf{v}_{\text{obs}}$ 
end for

```

5.2.1 Generating an Unconditional Velocity Field, $\mathbf{V}$

To generate $\mathbf{V}$ efficiently based on the assumptions in Section 5.1 we use a technique involving the 2-dimensional discrete Fourier transform, which for a matrix $\mathbf{M} \in \mathbb{R}^{M \times N}$ is given by

$$\mathcal{F}[\mathbf{M}] = \left\{ \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \mathbf{M}_{m,n} \exp \left\{ -2\pi i \left(\frac{mj}{M} + \frac{nk}{N} \right) \right\} \right\}_{j,k=0}^{M-1, N-1}. \quad (20)$$

We first draw a sample $\mathbf{w}$ of size MN from the standard normal distribution, rearrange it to a matrix $\mathbf{W}$ of size $M \times N$, and use the result that

$$\mathbf{V} = \mathcal{F}[\mathcal{F}[\boldsymbol{\Sigma}]^{1/2} \odot \mathcal{F}^{-1}[\mathbf{W}]] + \boldsymbol{\mu} \quad (21)$$

is the matrix-form of a sample from the prior distribution, $N(\boldsymbol{\mu}, \boldsymbol{\Sigma}_T)$ (Chan and Wood, 1997). Here, $\odot$ denotes element-wise multiplication, $\boldsymbol{M}^{1/2}$ denotes the square root of each element in $\boldsymbol{M}$, and $\mathcal{F}^{-1}$ is the inverse 2-dimensional discrete Fourier transform.

In Figure 24 we display a part of a velocity field generated by this method. The velocities at points that are close together are reasonably correlated, indicating that the correlation parameter $\phi = 10$ is a good choice. Note also that the velocities generally take values in a range from 0 to 2 m/s, and as such satisfy the criterion we set for reasonable paths in Section 4.4.

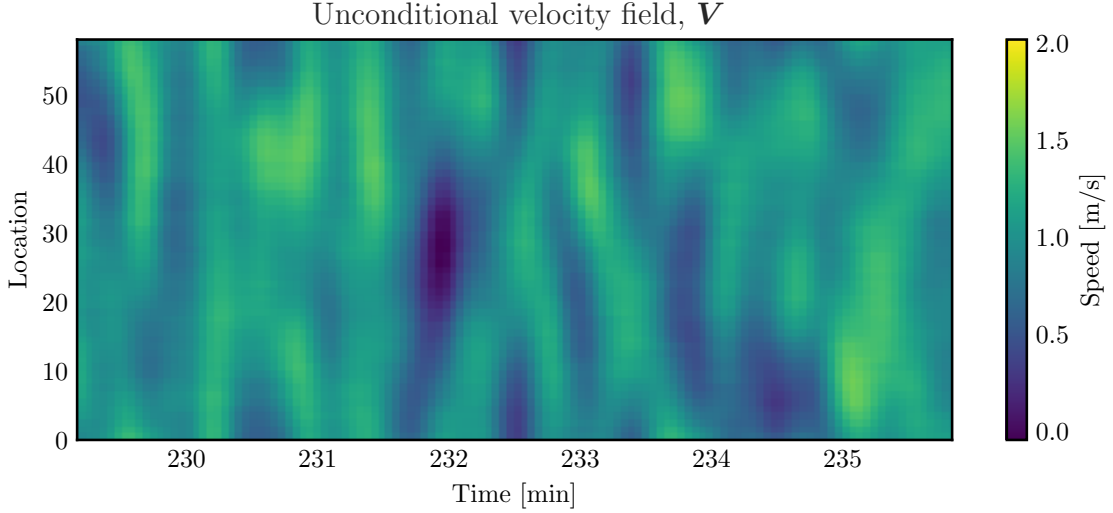


Figure 24: Velocity field modeled as a Gaussian random field with $\boldsymbol{\mu} = \mathbf{1}$ and covariance matrix $\boldsymbol{\Sigma}_T$.

5.2.2 Conditioning $\mathbf{V}$ on Observations

Once we have a velocity field $\mathbf{V}$, we generate a conditional velocity field $\tilde{\mathbf{V}}$ based on the current observations stored in $\mathbf{A}_{\text{obs}}$ and $\mathbf{v}_{\text{obs}}$. To perform the conditioning we employ a method described by Rue and Held (2005) and Eidsvik et al. (2009). We outline the necessary calculations here, but for more in-depth theory we refer to the literature. Under the assumptions stated in Section 5.1 we draw a sample $\tilde{\mathbf{v}}$ from the posterior distribution using

$$\tilde{\mathbf{v}} = \mathbf{v} + \mathbf{u}, \quad \mathbf{u} = \boldsymbol{\Sigma}_T \mathbf{h}, \quad \mathbf{h} = \mathbf{A}'_{\text{obs}} [\mathbf{A}_{\text{obs}} \boldsymbol{\Sigma}_T \mathbf{A}'_{\text{obs}} + \mathbf{P}]^{-1} (\mathbf{v}_{\text{obs}} - \mathbf{A}_{\text{obs}} \mathbf{v}). \quad (22)$$

If we define $\mathbf{H}$ and $\mathbf{U}$ as the matrix forms of $\mathbf{h}$ and $\mathbf{u}$, we can apply a variation of (21), and calculate

$$\mathbf{U} = \mathcal{F} [\mathcal{F}[\boldsymbol{\Sigma}]^{1/2} \odot \mathcal{F}^{-1}[\mathbf{H}]]. \quad (23)$$

Using this, we obtain the conditional velocity field $\tilde{\mathbf{V}} = \mathbf{V} + \mathbf{U}$.

As mentioned earlier, the purpose of conditioning the velocity field on observations is to create a velocity field that incorporates information from the reasonable DTW

lines. Then, by simulating paths based on $\tilde{\mathbf{V}}$, we generate lines that reflect the observed paths, potentially producing reasonable lines where the DTW model failed. To start the iteration, we need a reasonable initial guess to replace the 53 failed paths, to ensure that the initial conditional velocity field does not suggest unrealistic walking speeds. In Section 4.4, we found that the walking speed of the parade fluctuates around 1 m/s, so a reasonable starting point is to replace the 53 failed paths with straight lines in the (s, t) -plane with a constant speed of 1 m/s. Using these lines as initial observations also “encourages” the model to find new paths that the DTW algorithm could not, as we illustrated in Figure 23. Since the initial observations will be updated iteratively to minimize loss, anyways, the most important aspect of the initial guess is that their slopes are reasonable.

In the left display of Figure 25 we show a velocity field conditioned on 87 paths. These are the 34 reasonable DTW paths and 53 straight lines corresponding to a velocity of 1 m/s. The right display shows the difference between the conditional and the unconditional field. We observe that they are quite similar, but with slight variations due to the observed velocities.

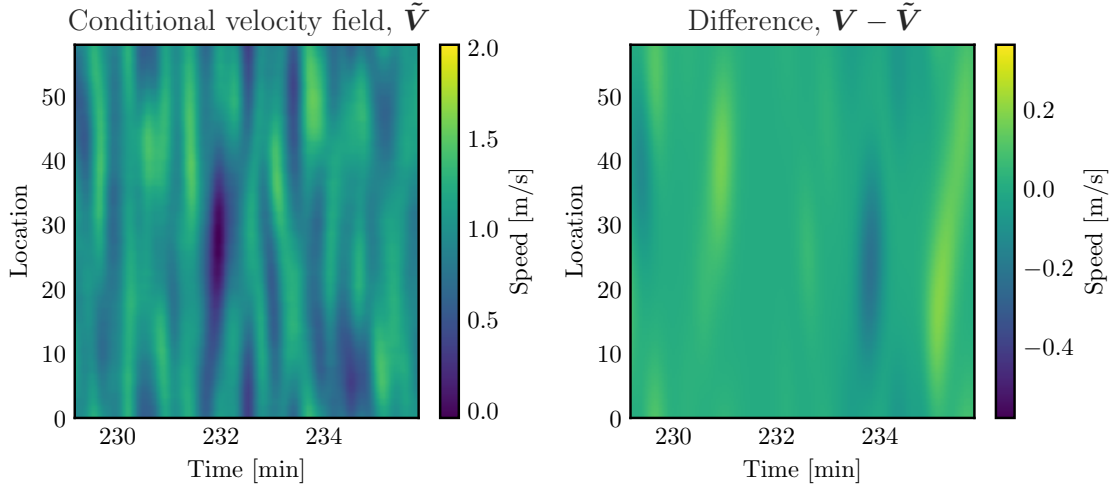


Figure 25: Conditional velocity field (left display) and difference between unconditional and conditional field (right display).

5.2.3 Simulating and Accepting New Observations based on $\tilde{\mathbf{V}}$

Using the conditional velocity field we simulate new paths. To do this we extract the time of entry of each part of the parade, i.e. the time of the 87 peaks shown in Figure 19, and extract velocities from the corresponding points in $\tilde{\mathbf{V}}$. Given these velocities we can calculate the time value when each group will reach the next location, extract the velocity from the corresponding point in $\tilde{\mathbf{V}}$, and continue until each path reach the final location. We store the new paths and corresponding velocities in $\mathbf{A}_{\text{new}}$ and $\mathbf{v}_{\text{new}}$.

We choose which of the simulated paths to accept based on total loss along each path. As in Section 4, we use cross-correlation for this purpose. For each line, we calculate the loss by walking along the path and summing up the alignment error

$1 - r(t, s, \delta)$ at each step. If one of the new simulated paths has lower loss than the corresponding path stored in $\mathbf{A}_{\text{obs}}$, then the new path is accepted to replace the previously observed path. We then update $\mathbf{A}_{\text{obs}}$ and $\mathbf{v}_{\text{obs}}$ to contain the coordinates of the new, improved paths and the corresponding velocities, and start the next iteration.

If the total change in loss of all paths is zero for several iterations in a row, say 5 iterations, we assume that the paths are optimized for the current velocity field. We then generate a new velocity field, condition on the last found paths, and continue the iteration. We do this for the sake of computational efficiency, even though the paths may not be fully optimized given the current velocity field.

5.3 Applying the Joint Velocity Warping Algorithm

Now we apply the algorithm with $B = 1000$ iterations to the DAS dataset from Oslo. First we analyze the simulated paths, then we discuss convergence.

5.3.1 Simulated Paths

In Figure 26 we show the paths obtained at the highest and lowest peaks in cross-correlation at the first location. At the highest peak the DTW path was reasonable to begin with, but here we simulate a path using the last conditional velocity field to compare performance with DTW. As we observe in the left display, the two methods provide similar paths, indicating that the conditional velocity field indeed captures information from the successful DTW path. In the right display, we observe that the joint method provides a more reasonable path than the DTW. By construction, the black line begins at the correct location, whereas DTW fails due to the weak signal for this particular group.

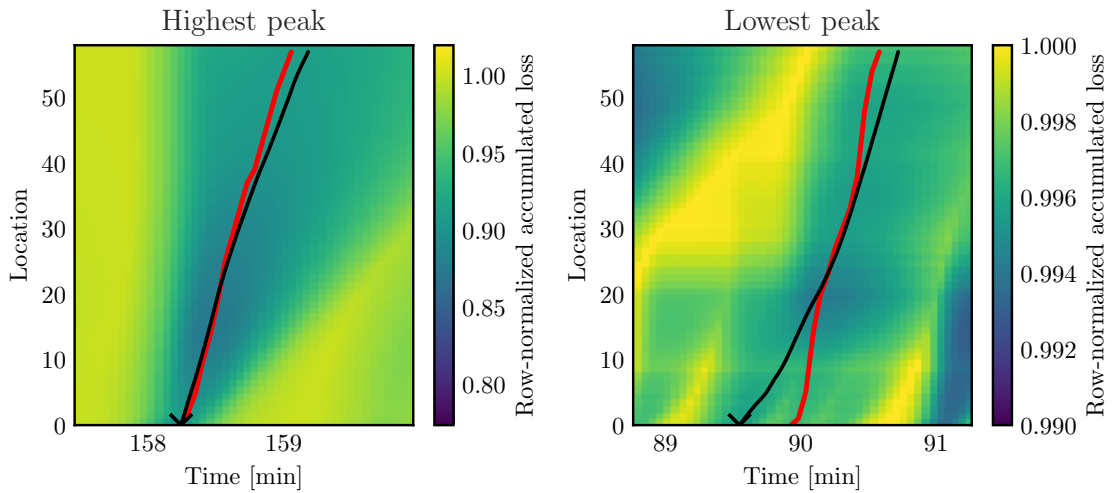


Figure 26: DTW paths (red) and joint velocity warping paths (black) found at the highest peak (left display) and the lowest peak (right display). The point of each peak is marked by a black cross.

Simulated paths at time steps of high activity are shown in Figure 27. The black lines found by the joint model appear to follow the DTW paths closely, and separate from them once they suggest unreasonable speeds. As such, these paths seem more appropriate as an estimate for tracking marching bands. Some lines, such as the last one, appear almost straight. In these cases it is difficult to say whether the model actually improves on the initial guess, or if guessing a diagonal line is just as good.

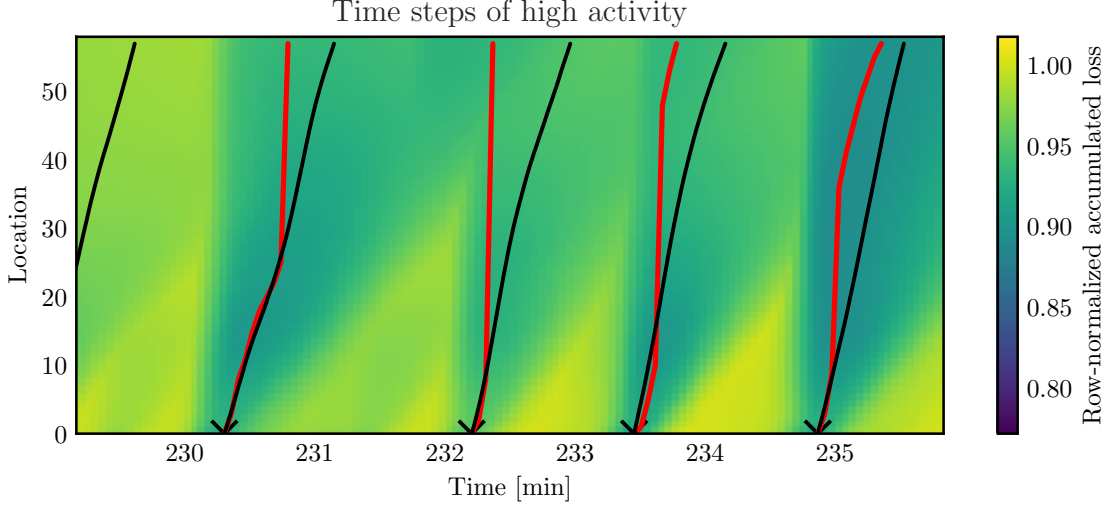


Figure 27: DTW paths (red) and joint velocity warping paths (black) at four consecutive peaks at time steps of high activity. The point of each peak is marked by a black cross.

Overall, the joint model outperforms the DTW-algorithm in time steps of high activity and when signals are weak. It must be mentioned that this partly is due to the assumptions we make in the model, which forces the simulated velocities to be within a reasonable range. This makes the model well fit to track marching bands in the May 17th case, causing the joint model to give better results. On the other hand, DTW relies solely on the cross-correlation, and is therefore more reliant on the accuracy of this similarity measure. Although this causes DTW to fail more often in our specific case, the lack of additional assumptions make it more applicable in general situations where we do not have as much information on the pedestrian movement to begin with.

5.3.2 Convergence

In the previous subsection we set the total number of iterations to $B = 1000$. Figure 28 displays the convergence rate over these iterations, illustrating the decrease in loss relative to the loss along the initial paths. The jagged movement of the loss is a result of the occasional updating of the initial velocity field, indicating that it is efficient for the sake of convergence to update the velocity field frequently. Although the algorithm do not appear to have converged completely, the loss flattens out quite substantially, and as such it is likely little to be gained by increasing B further.

After 1000 iterations, the loss relative to the loss along the 87 initial lines reaches

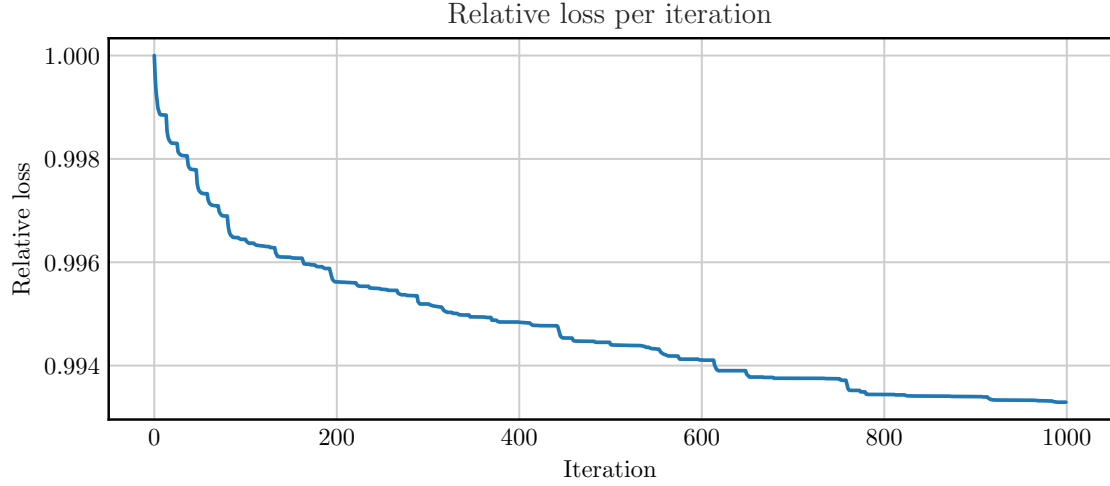


Figure 28: Relative loss per iteration in the joint velocity warping algorithm.

0.993 (Figure 28). For comparison, the total loss calculated along the 87 DTW lines is approximately 0.986 of the initial displayed loss. This means that the joint velocity warping algorithm gives higher loss compared to DTW. The reason for this is that DTW is designed specifically to minimize the loss along its path, while the joint model is constructed to minimize loss while also being a realistic sample from a conditional velocity field. This causes the cross-correlation along the paths to be slightly off. This is, however, not necessarily a negative thing, considering that the cross-correlation is not a perfect measure of the location of the parade, as shown by the failed DTW lines. Still, it shows that the particular cases we illustrated in Figure 23 are not sufficient to simulate lines with lower overall loss than the DTW lines. Nonetheless, the lines provided by the joint velocity warping model are more reasonable than the failed DTW-lines, and at the same time the loss is lower than that of a trivial guess of 1 m/s constant speed. As such, the model is successful in generating a reasonable tracking of the marching bands in the parade.

6 Conclusion

The primary objective of this project was to track marching bands in the 17th of May parade in Oslo by the use of DAS data. To achieve this, we began with data analysis, developed a new version of the DTW algorithm, and identified 87 potential marching bands using peaks in cross-correlation. We then attempted to track these bands along Karl Johans gate. Our findings show that the adapted DTW model effectively identifies paths with reasonable walking speeds when events are isolated and the signal strength is powerful enough, but it struggles with overlapping events and weak signals. In total, DTW gave 34 reasonable paths, while the remaining 53 suggested walking speeds exceeding 3 meters per second.

In an attempt to improve the 53 paths we created the joint velocity warping model, which models the velocity of the parade as a Gaussian random field. The method generated paths that were similar to the 34 successful DTW paths, and significantly improved the 53 unsuccessful ones. These paths were more reasonable than assuming constant speed of the parade, and as such the methods were successful in tracking marching bands. This demonstrates the potential of DAS technology as a scalable, non-intrusive solution for monitoring pedestrian movements at large public events, where there is a prior knowledge on the movement patterns of the pedestrians.

Monitoring pedestrian flow in urban areas with DAS has promising outlooks, but requires more research. Our main struggle was finding a similarity measure that effectively distinguishes between different groups of pedestrians, which is necessary if DTW is to find reasonable paths. We found cross-correlation between PSDs to be the most accurate measure, but discovering other alternatives is a relevant topic for future work. For instance, incorporating machine learning models for this purpose could provide more robust and adaptive similarity measures, particularly in cases with overlapping events or weak signals. In addition to this, the use of cameras or other visual inspection methods to validate the DAS data against model outputs could improve the interpretability of the results, which would allow for a more thorough assessment of the model performance. Another potential next step is to apply velocity warping to less controlled environments to evaluate its robustness and suitability for monitoring pedestrians in more general situations.

Bibliography

- Abrahamsen, P. (1997). *A review of gaussian random fields and correlation functions*. Norsk Regnesentral.
- Berndt, D. J., & Clifford, J. (1994). Using dynamic time warping to find patterns in time series. *Proceedings of the 3rd international conference on knowledge discovery and data mining*, 359–370.
- Brownjohn, J. M., Pavic, A., & Omenzetter, P. (2004). A spectral density approach for modelling continuous vertical forces on pedestrian structures due to walking. *Canadian Journal of Civil Engineering*, 31(1), 65–77.
- Chan, G., & Wood, A. T. (1997). Algorithm as 312: An algorithm for simulating stationary gaussian random fields. *Applied Statistics*, 171–181.
- Efrat, A., Fan, Q., & Venkatasubramanian, S. (2007). Curve matching, time warping, and light fields: New algorithms for computing similarity between curves. *Journal of Mathematical Imaging and Vision*, 27(3), 203–216.
- Eidsvik, J., Martino, S., & Rue, H. (2009). Approximate bayesian inference in spatial generalized linear mixed models. *Scandinavian journal of statistics*, 36(1), 1–22.
- Glasbey, C. A., & Mardia, K. V. (1998). A review of image-warping methods. *Journal of applied statistics*, 25(2), 155–171.
- Hale, D. (2013). Dynamic warping of seismic images. *Geophysics*, 78(2), S105–S115.
- Harris, F. J. (1976). *Windows, harmonic analysis, and the discrete fourier transform*. Naval Undersea Center.
- He, Z., & Liu, Q. (2021). Optical fiber distributed acoustic sensors: A review. *Journal of Lightwave Technology*, 39(12), 3671–3686. <https://doi.org/10.1109/JLT.2021.3059771>
- Johannessen, K., Drakeley, B., & Farhadiroushan, M. (2012). Distributed acoustic sensing-a new way of listening to your well/reservoir. *SPE intelligent energy international conference and exhibition*, SPE-149602.
- Kreyszig, E. (2011). *Advanced engineering mathematics*. John Wiley & Sons.
- Landau, H. (1967). Sampling, data transmission, and the nyquist rate. *Proceedings of the IEEE*, 55(10), 1701–1706.
- MapTiler. (2024). Retrieved September 10, 2024, from <https://cloud.maptiler.com/maps/91c5d910-0d0f-4d98-814f-92026a393b25/>
- Martani, C., Stent, S., Acikgoz, S., Soga, K., Bain, D., & Jin, Y. (2017). Pedestrian monitoring techniques for crowd-flow prediction. *Proceedings of the Institution of Civil Engineers-Smart Infrastructure and Construction*, 170(2), 17–27.
- Mateeva, A., Lopez, J., Potters, H., Mestayer, J., Cox, B., Kiyashchenko, D., Wills, P., Grandi, S., Hornman, K., Kuvshinov, B., et al. (2014). Distributed acoustic sensing for reservoir monitoring with vertical seismic profiling. *Geophysical Prospecting*, 62(4-Vertical Seismic Profiling and Microseismicity Frontiers), 679–692.
- Ratanamahatana, C. A., & Keogh, E. (2004). Everything you know about dynamic time warping is wrong. *Third workshop on mining temporal and sequential data*, 32.

-
- Rørstadbotnen, R. A., Eidsvik, J., Langhammer, J., Landrø, M., & Ibrahim, O. (2023a). Using fiber optic communication network to monitor urban activity. *Available at SSRN 4666629*.
- Rørstadbotnen, R. A., Eidsvik, J., Bouffaut, L., Landrø, M., Potter, J., Taweestintananon, K., Johansen, S., Storevik, F., Jacobsen, J., Schjelderup, O., et al. (2023b). Simultaneous tracking of multiple whales using two fiber-optic cables in the arctic. *Frontiers in Marine Science*, 10, 1130898.
- Rue, H., & Held, L. (2005). *Gaussian markov random fields: Theory and applications*. Chapman; Hall/CRC.
- Sauer, T. (2014). *Numerical analysis*. Pearson Education Limited.
- United Nations Human Settlements Programme. (2013). *Planning and design for sustainable urban mobility: Global report on human settlements 2013*. Routledge.
- Zhang, T.-Y., Shi, B., Zhang, C.-C., Xie, T., Yin, J., & Li, J.-P. (2021). Tunnel disturbance events monitoring and recognition with distributed acoustic sensing (das). *ISRM International Symposium-Asian Rock Mechanics Symposium*, ISRM-ARMS11.
- Zhu, H.-H., Liu, W., Wang, T., Su, J.-W., & Shi, B. (2022). Distributed acoustic sensing for monitoring linear infrastructures: Current status and trends. *Sensors*, 22(19), 7550.
- Zhu, T., Shen, J., & Martin, E. R. (2020). Sensing earth and environment dynamics by telecommunication fiber-optic sensors: An urban experiment in pennsylvania usa. *Solid Earth Discussions*, 2020, 1–30.
- Živanović, S., Pavić, A., & Ingólfsson, E. T. (2010). Modeling spatially unrestricted pedestrian traffic on footbridges. *Journal of structural engineering*, 136(10), 1296–1308.